

The Ecological Inequality Trap

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Abstract

This paper develops a theory of biological resource extraction, where wealth inequality jointly evolves with ecological dynamics. It features an equilibrium *ecological inequality trap* where the rich operate capital-intensive technology to extract high-value resources while the poor rely on labor-intensive capture of alternative resources. When the initial wealth distribution is highly unequal, the laissez-faire economy falls into this trap while a constrained social planner reaches an egalitarian steady state, by transiently suppressing the rich's resource and conserving the poor's. The framework is applied to Lake Victoria, where the invasive Nile perch was introduced in the 1960s to commercialize the fishing industry but caused severe biodiversity loss of native species. The results highlight the efficiency–equity tradeoff of biodiversity conservation policies central to the current debate: one-sided protection of certain species may work against the economic interests of those who depend on them, even in the long run.

JEL Classification: Q22, Q57, O13, H23

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1 Introduction

Biodiversity is changing more rapidly than ever before, yet its economic and distributional consequences remain largely under-explored. As Dasgupta (2021) observes, the costs of ecological degradation may fall disproportionately on poorer households with fewer margins of adaptation. This raises two important questions. The first, positive question concerns how ecological dynamics shape economic inequalities when heterogeneous households differ in their capacity to exploit specific resources and adapt to ecological changes; and how the resulting inequalities feed back into the ways natural resources are being extracted, and in turn, future ecological change (Barrett et al., 2011). The second is normative: how should optimal biodiversity conservation policy, famously framed by Weitzman (1998) as the “Noah’s Ark” problem, incorporate these distributional considerations? This paper puts forward a theoretical framework of natural resource extraction, where ecological dynamics and wealth inequality jointly evolve, to examine these questions.

We motivate the theory with Lake Victoria in Africa. Employing over three million people in fisheries-related activities, the lake has undergone one of the most dramatic documented episodes of biodiversity loss in the past century, making it an unusually sharp setting for our theory (Kaufman, 1992; Nyamweya et al., 2023). Nile perch were introduced into the lake in the mid-twentieth century as part of an effort to expand commercial fisheries and raise fishery revenue.¹ The intervention initially appeared to succeed. The boom of Nile perch, once regarded as the “Saviour fish,” transformed the local fishing economy, generating large income gains through exports to global markets and bringing local population to enter the fishing industry (Weston, 2022). Yet these gains came with substantial ecological costs. As a large predatory species, Nile perch contributed to the collapse of many endemic haplochromine cichlids, a group of smaller fish that had long supported local consumption and lower-value fisheries, also known as the “trash fish” (Goldschmidt et al., 1993; Kitchell et al., 1997; Pringle, 2005a). In recent decades, however, the Nile perch fishery has faced mounting pressure from overfishing and pollution from surrounding catchments, with catches declining after the mid-2000s. At the same time, some haplochromine populations and other small pelagic fish have shown signs of recovery, with their landings increasing as predation pressure from Nile perch has weakened (Balirwa et al., 2003).

However, what makes the Lake Victoria case even more informative is that it is not only an ecological system, but also an economy in which households differ in their ability to respond to ecological change. Nile perch is a high-value species, but harvesting it requires capital: motorized vessels, higher-quality gillnets and longlines, and the ability to travel farther from shore (Pringle, 2005a; Mpomwenda et al., 2022). Access to the Nile perch boom was therefore shaped by pre-existing wealth distribution. Richer fishermen could invest in Nile perch catch technology, while poorer fishermen remained more dependent on lower-value, labor-intensive species, including haplochromines and other small fish consumed locally (Beuving, 2010). Using DHS data and publicly available data from the Lake Victoria Fisheries Organization (LVFO), we document that during the Nile perch boom, wealth inequality rose among fishing households, with capital deepening concentrated among Nile perch fishermen. Furthermore, as Nile perch catches declined in later years, species specialization diverged, with some fishermen remaining in Nile perch while others shifted again toward lower-value species. These facts point to an economic–ecological feedback²: ecological shocks change the distribution of income

¹See Pringle (2005b) for a historical account of the introduction of Nile perch into Lake Victoria.

²Similar feedbacks arise in other resource-dependent settings: ecological change reallocates returns across species and technologies, while households’ ability to respond depends on their financial, social, and human capital (Cinner et al., 2015, 2018; OECD, 2011; Barange et al., 2018). Furthermore, unequal access to productive technology can reinforce inequality and resource degradation, generating poverty–environment traps (Mirza et al., 2019; Barrett et al., 2011).

by reallocating returns across fishing technologies, while wealth inequality shapes which technologies households adopt and therefore which species face harvesting pressure. Our framework study this feedback and ask how conservation policy should be designed when biodiversity, technology choice, and the distribution of wealth evolve jointly.

We develop a dynamic model featuring the predator-prey dynamics in the ecological system, where each economic group decide on how to allocate labor and capital across a labor-intensive prey fishery and a capital-intensive predator fishery, and how much to consume and save out of its harvesting income. We characterize both the laissez-faire equilibrium and the constrained social optimum, along their full transition paths and at their steady states. The two differ because private agents harvest at market prices, ignoring that their catch today depletes a shared stock tomorrow and, through the predator-prey link, alters the productivity of the *other* group’s fishery; the planner instead harvests at social shadow values that internalize both channels. Due to the minimum capital requirement of the Nile perch catch technology, the economy admits multiple steady states. In an *egalitarian* steady state, both groups operate the capital-intensive Nile perch fishery and their wealth gradually converges. In what we term an *ecological inequality trap*, the poor are confined to the low-value prey fishery while the rich operate the predator fishery, and the ecological state and the wealth distribution jointly sustain this specialized fishing choice. We further derive the policy that decentralizes the constrained optimum: a system of species- and group-specific Pigouvian harvest taxes that price each catch at its social marginal value, correcting the agents’ failure to internalize the stock and cross-species externalities.

We calibrate the model to the relative beach prices of the major fish species and to the predator-prey dynamics of Lake Victoria: we draw the ecological block from the bio-economic estimates of [Brown et al. \(2005b\)](#) and discipline the harvest technology parameters with the empirical fisheries-production literature. We then use the calibrated model to characterize the efficiency and distributional consequences of the biodiversity changes.

Our quantitative results suggest that, first, relative to the constrained optimum, the inequality trap under laissez-faire sustains *lower* stocks of both species: the poor over-harvest the prey on which they depend, while the unmanaged predator boom further depresses it. Lower prey stock also leads to lower predator stock compared to the socially optimal level. Second, and more importantly, we map the entire space of initial wealth distribution of the two groups into the steady state each economy evolves into, and characterize three regions. When the poor and the rich both begin wealthy enough, the laissez-faire and the planner reach the egalitarian steady state; when the two groups both begin relatively poor, both regimes are resigned to the inequality trap, because, without redistributing wealth, the constrained planner is not able to engineer an escape when the rich themselves are relatively lower capitalized. Between these lies a region of intermediate aggregate wealth and *high initial inequality* in which the laissez-faire and the social optimal paths diverge: the planner lifts the poor out of the inequality trap but the laissez-faire remains in the trap. The reason the disagreement appears is that a well-capitalized rich group earns enough from fishing Nile perch that the planner finds it cheap to rebalance harvesting in the poor’s favor, i.e., suppressing the booming predator and conserving the prey, so that the poor’s own income rises enough to carry it across the capital threshold. The escape is implemented through the Pigouvian formulas (without any transfer of wealth between groups): by repricing the two catches, the planner raises the poor’s prey income, finances its accumulation past

For example, [McNally et al. \(2011\)](#) show that mangrove protection in Tanzania generated short-run losses in mangrove firewood consumption, especially among richer households, while all wealth classes appear to benefit from longer-run gains in shrimping and fishing from improved ecosystem sustainability.

the threshold, and brings it into the perch fishery.

We evaluate the welfare loss of the laissez-faire economy relative to the social optimum aggregated over the transition path. We find substantial welfare loss for the region of initial wealth distribution where the laissez-faire and the social planner disagree on the steady state. This comes from the fact that the poor are permanently excluded from the high-productivity technology under the former regime, and the welfare loss therefore captures the forgone escape from the inequality trap. Surprisingly, there is also large aggregate welfare loss when both groups begin wealthy: both operate the perch fishery at scale throughout the boom, which leads to the greatest harvesting pressure and hence the externality that laissez-faire fails to internalize.

Finally, we evaluate two alternative policy instruments motivated by the Lake Victoria policy debate³, applied to an initial condition leading to the inequality trap. The first is a minimum mesh-size regulation, a central technical rule in the management of Lake Victoria fisheries (Lake Victoria Fisheries Organization, 2015, 2016, 2021). The second is a rights-based access instrument, which we model as a tradable quota; while current Lake Victoria policies emphasize user rights, permits, and effort controls rather than a fully developed ITQ system, property-rights approaches have been proposed in the Lake Victoria literature and are central to broader fisheries policy debates (Brown et al., 2005a; Costello et al., 2008). We show that the minimum mesh size can backfire: because it acts as an untargeted tax on whoever harvests the protected species, protecting the prey falls on the poor and widens inequality, while protecting the predator falls on the rich and narrows inequality, in each case working against the very group the conservation ostensibly serves, and reducing aggregate welfare throughout. A tradable quota, by contrast, traces an inverted-U in aggregate welfare, and over a range of intermediate level quota it does *better* than the constrained social optimum itself. There are two forces behind this: the quota restricts the perch catch and so, like the mesh size regulation, protects the predator at the prey’s expense, an ecological cost rising in stringency of the quota; but because permits are allocated equally and the excluded poor sell theirs to the rich, the quota also redistributes resource rents, a channel the constrained planner lacks, and at intermediate range of quotas this redistribution effect dominates. Yet the rents do not free the poor from the trap: the same tightening that raises the rent also lowers the return to operating the perch fishery, so the poor consumes the windfall rather than investing it. Distributional considerations thus turns out to be central to conservation policy in this setting: which species one protects, and through which instrument, determines who bears the cost and who, if anyone, escapes from the inequality trap.

Related Literature This paper is related to the following strands of literature. First, the economic drivers and consequences of biodiversity change. Recent work studies biodiversity as productive natural capital and quantifies the economic costs of ecosystem disruption. Giglio et al. (2024) develop a macroeconomic model in which natural species provide ecosystem services that complement production; Liang et al. (2025) estimate the relationship between economic activity and biodiversity in the United States. Frank and Sudarshan (2024) and Frank (2024) provide causal evidence that the collapse of keystone species can generate large social costs through health, sanitation, and agricultural production. We contribute to this literature by understanding a two-way feedback between biodiversity and inequality, where species are productive assets used by heterogeneous households.

Second, the paper contributes to the literature on optimal biodiversity conservation. Weitzman (1998) formalizes the classic “Noah’s Ark” problem as a cost-effectiveness problem of choosing which

³Our policy simulations abstract from enforcement constraints. In practice, enforcement is itself a central constraint in Lake Victoria fisheries governance: for example, limited administrative capacity, political interference, and corruption may undermine compliance with gear restrictions and access regulations (Nunan et al., 2018; Nunan, 2020).

species to protect, while Brock and Xepapadeas (2003) provide an insurance argument for valuing biodiversity and designing conservation priorities. We also speak to the distributional implications of rights-based fisheries management, which is often regarded as one of the core policy to prevent fishery collapse (Costello et al., 2008). Our departure is to introduce household heterogeneity and endogenous resource extraction into this problem. In our model, the value of protecting a species depends endogenously on which households derive income from it and we discuss how conservation changes the distribution of wealth. Ecosystem interactions create a link between efficiency and equity: protecting one species changes the productivity of other species, thereby affecting the returns to the technologies used by different groups. Optimal conservation policy therefore requires accounting for both ecological externalities and distributional consequences.

Third, the paper relates to the theory of common-pool resource extraction and to models of poverty and inequality traps. The classic literature on fisheries and renewable resources studies open-access extraction and socially optimal management (Gordon, 1954; Dasgupta and Heal, 1979; Levhari and Mirman, 1980). Development theory shows how capital-market imperfections, non-convex technologies, and occupational choice can make the initial distribution of wealth persistently affect long-run outcomes: for example, Galor and Zeira (1993), Banerjee and Newman (1993) and more recently Ghatak and Newman (2025). We combine these two ideas. The common resource is embedded in a multi-species ecological system, and agents differ in their ability to access the high-return, capital-intensive extraction technology⁴. Our inequality trap is therefore generated through ecological interactions: we introduce heterogeneous-agent dynamics into a common-pool resource environment and shows how ecological change can become a source of persistent inequality.

We organize the paper as follows: Section 2 provides motivating empirical facts on Lake Victoria. Section 3 presents the core theoretical framework and describe the calibration process and solution algorithm. Section 4 present the quantitative results. Section 5 concludes.

2 Background and Facts

This section sets out a few motivating facts of the Lake Victoria fishery that motivate our model. The fishery has undergone an ecological regime shift in a harvested predator–prey system; the species in that system are sold with different prices; the technology used to capture the more valuable of them is capital-intensive and unevenly held; and the dispersion of wealth among fishing households is rising as communities diverge in the species they pursue. Together these describe a human–ecological system in which heterogeneous, wealth-constrained fishers exploit a shared and depleting resource.

Aggregate Capture Over Time Figure 1 summarizes the transformation of the Lake Victoria fishery over the past seven decades based on total capture. Panel (A) plots total capture production by species for the three riparian states. Through the 1970s the fishery industry was modest and broadly based a few native species, but the introduction of Nile perch (*Lates niloticus*) triggered an abrupt regime shift: perch landings rose from under 80,000 tonnes around 1980 to a peak of roughly 372,000 tonnes by 1990, at which point they accounted for some 60% of the recorded catch. The expansion coincided with the near-disappearance of the native haplochromine cichlids from the landings. Since the 1990s the composition has shifted again, as the small pelagic dagaa (silver cyprinid, *Rastrineobola argentea*) has grown to overtake Nile perch in volume, supplying close to 40% of the

⁴One of the most related papers is Mirza et al. (2019) which put forward a framework that features wealth accumulation allowing households to take up more productive technology to capture natural resources, but this may depress the growth of resources. We allow for household heterogeneity in capturing different types of resources and and ecosystem interactions of these resources; and discipline our model by the empirical patterns in Lake Victoria fisheries.

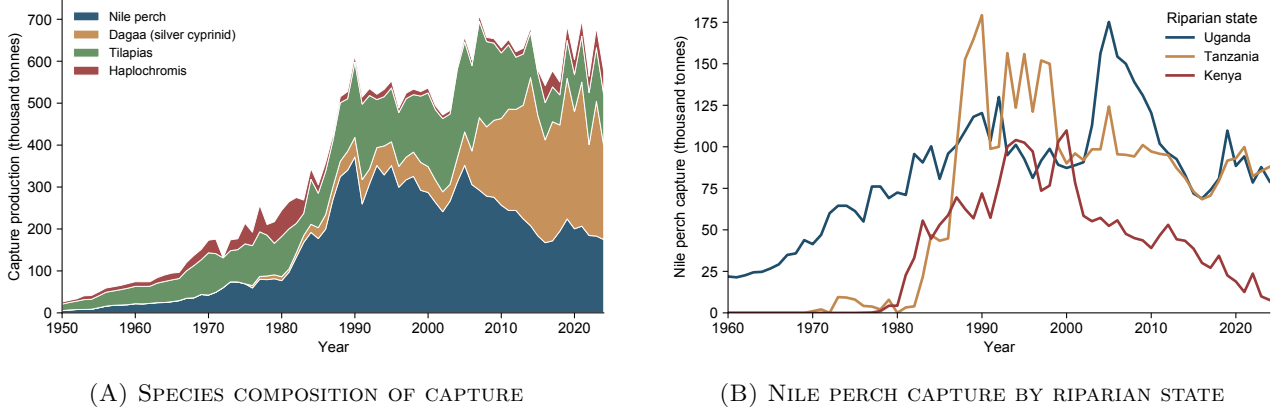


FIGURE 1: LAKE VICTORIA FISHERIES: CAPTURE PRODUCTION, 1950–2024

Notes: Panel (a) shows total capture production by species for the three riparian states (Kenya, Tanzania, and Uganda) combined; “Tilapias” aggregates Nile tilapia (*Oreochromis niloticus*) and other tilapias *nei*, and “Haplochromis” corresponds to the FAO category mouthbrooding cichlids (*Haplochromis* spp.). Panel (b) shows Nile perch (*Lates niloticus*) capture by riparian state. Series are aggregated from national capture reported in the FAO statistical area *Africa–Inland waters* and are used as a proxy for Lake Victoria; the Tanzanian series additionally includes minor catches from other inland waters. Dagaa (silver cyprinid, *Rastrineobola argentea*) is under-recorded prior to 1980.

Source: FAO, Global Capture Production (release 2026.1.0). <https://www.fao.org/fishery/static/Data>

catch by 2024 while perch has receded to about 30%. The haplochromines, in turn, have also staged a partial resurgence in more recent years, with landings recovering from around 12,000 tonnes through the 2000s to some 55,000 tonnes by 2024. Panel (B) decomposes Nile perch landings by riparian state: after the Nile perch catch peak, Kenyan landings collapse toward zero, whereas Uganda and Tanzania sustain substantial harvests under continued fishing effort.

The trajectory in Panel (A) traces out a predator–prey interaction that motivates our model. In early stages, the Nile perch, i.e., a large piscivore, expanded explosively and preyed down the native haplochromine cichlids, whose collapse from the landings mirrors the perch boom almost exactly; the small pelagic dagaa, released from competition with the haplochromines, expanded into the vacated niche. However, later on, the predator itself was checked: intense and largely open-access fishing pressure, compounded by deteriorating water quality in the lake, drove Nile perch landings down from their 1990 peak. As predation pressure relaxed, the smaller species recovered, i.e., haplochromine catches climbing back and dagaa rising to overtake perch in volume. The data thus describe a coupled system in which predator and prey rise and fall in opposition, with human harvesting acting on both, precisely the structure the model formalizes.

Beach Prices One important feature of the species in this system is the hierarchy of their values which gives the regime shift its bite for fishers’ livelihoods. Figure 2 reports the implied beach (ex-vessel) price of the four species central to our analysis. For Kenya we draw on the annual *Fisheries Statistical Bulletins* (2011–2023); for Uganda, on the Lake Victoria Fisheries Organization *Catch Assessment Surveys* Information (2006–2011). In each source we compute the implied unit price as total landed value divided by total landed quantity in a given year. To make the two countries comparable, we express each price per kilogram in local currency, convert to US dollars at period-average official exchange rates, and deflate to constant 2023 dollars using the US consumer price index.⁵

Three features stand out. First, there is a pronounced and persistent *price hierarchy* across species.

⁵Exchange rates are World Bank WDI series `PA.NUS.FCRF` (period-average official rate, local currency per US\$); the deflator is the U.S. Bureau of Labor Statistics CPI-U for all urban consumers. Implied prices are value-weighted lake-wide averages and therefore reflect ex-vessel (beach) values rather than retail or export prices.

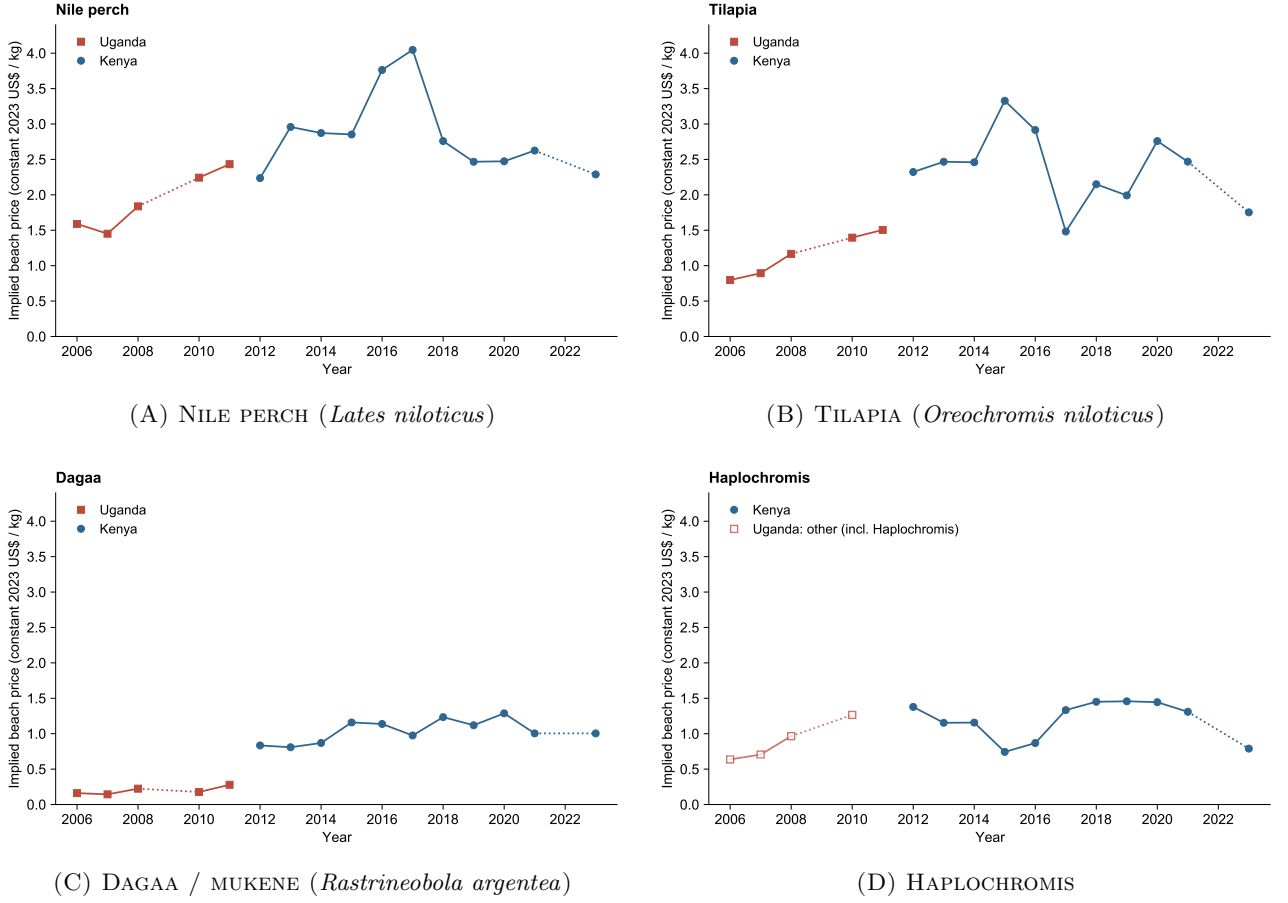


FIGURE 2: REAL AVERAGE BEACH PRICES OF MAJOR LAKE VICTORIA SPECIES, 2006–2023

Notes: Each panel plots the average beach (ex-vessel) price of one species, computed as annual landed value divided by annual landed quantity. The Ugandan series (2006–2011, terracotta squares) are drawn from the Lake Victoria Fisheries Organization Catch Assessment Surveys for the Ugandan waters of the lake; the Kenyan series (2012–2023, blue circles) are drawn from the Kenya Fisheries Annual Statistical Bulletins. All prices are first expressed per kilogram in local currency, converted to US dollars at period-average official exchange rates (World Bank WDI, series PA.NUS.FCRF), and then deflated to constant 2023 US dollars using the US consumer price index (BLS CPI-U, all urban consumers). Dotted segments interpolate across years with no survey (Uganda 2009; Kenya 2022). The 2017 Kenyan Nile perch price is replaced by the 2016–2018 average to smooth the outlier. In panel (d), Uganda does not report *Haplochromis* as a separate category; the hollow terracotta markers therefore show the aggregate “other fish species” group (which includes *Haplochromis*). All four panels share a common vertical scale.

Nile perch is by far the most valuable, trading on the order of US\$2.5–3 per kilogram in recent years and peaking close to US\$4 in the mid-2010s. Tilapia follows at roughly US\$2 per kilogram. The two small forage species, i.e., dagaa (*omena*/mukene) and haplochromis, are the cheapest, at around US\$1 per kilogram or below. These low-value species are precisely the “trash fish” on which the poorer fishing households disproportionately rely, and their low unit value relative to the predatory Nile perch is a central ingredient of the mechanism we will model.

Second, these real prices are *broadly stable over time*. None of the four price series displays a strong secular trend: each fluctuates within a fairly narrow band over nearly two decades, with the most visible movement being a transitory rise in the Nile perch price in the mid-2010s rather than any sustained drift. This stability is convenient for our calibration, since it suggests that a constant relative-price wedge between predator and prey species could be a reasonable approximation.

Third, the Kenyan and Ugandan series, although they are from different data sources, are *largely consistent in levels*. For the high-value species: at the 2011–2012 handoff between the two data sources, the Nile perch price is almost continuous (about US\$2.4 in Uganda in 2011 against US\$2.2 in Kenya in 2012), and tilapia lines up similarly; the agreement is similar for the cheapest species.

Overall, the concordance of two independent national statistical systems, in different currencies, lends credibility to the price levels we use.

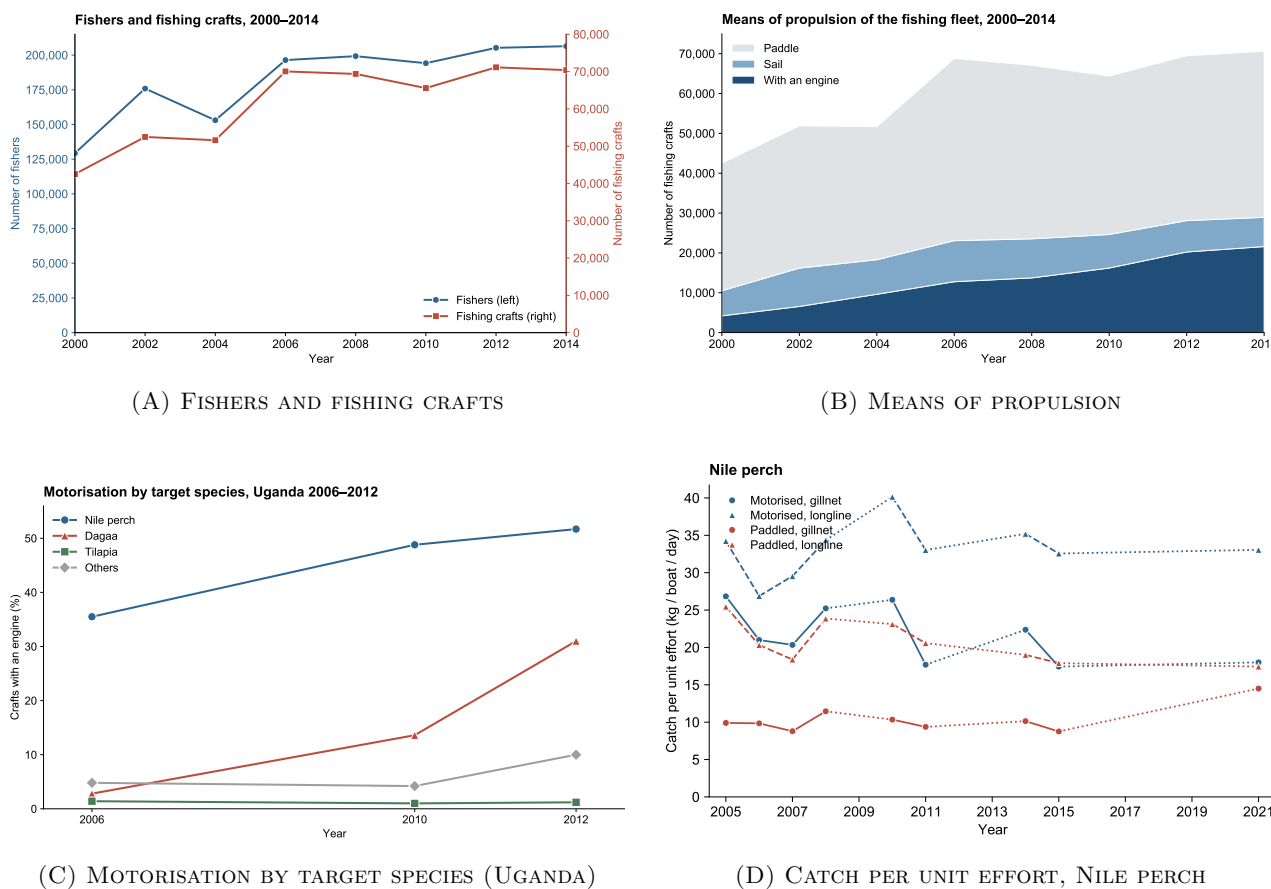


FIGURE 3: FISHING EFFORT AND PRODUCTION TECHNOLOGY

Notes: Panels (A)–(C) are constructed from the Lake Victoria Fisheries Organization (LVFO) Frame Survey Reports; panel (D) from the LVFO Catch Assessment Survey (CAS). Panels (A) and (B) plot lake-wide totals (Kenya, Tanzania, and Uganda) from the Regional Frame Survey Report 2014. Panel (A) shows the number of fishers (left axis) and fishing crafts (right axis). Panel (B) decomposes the fleet by means of propulsion (number of crafts); “with an engine” combines outboard and inboard engines. Panel (C) reports, for the Ugandan portion of the lake only, the share of crafts equipped with an engine by target species, computed as the number of engine-powered crafts targeting a species divided by all crafts targeting that species, from the Uganda Frame Survey Reports of 2006, 2010, and 2012; “Others” aggregates *Clarias*, *Protopterus*, *Synodontis*, and haplochromines. Panel (D) plots catch per unit effort (CPUE, kilograms per boat per day) for Nile perch by vessel-and-gear technology, from the CAS report 2021; survey-wave observations are first averaged across the three riparian countries (simple mean) and then averaged within each calendar year, with dotted segments spanning years in which no survey was conducted. “Motorized” and “paddled” denote Sesse vessels with and without an outboard engine, and “gillnet”/“longline” the gear deployed.

Fishing effort and production technology The predatory Nile perch is the most valuable fish in the lake; but more importantly, the capacity to capture it is not evenly distributed. Figure 3 characterizes the evolution of fishing effort on Lake Victoria and the technology with which it is deployed.

Panel (A) reports the total number of fishers and fishing crafts from the 2014 regional Frame Survey Reports. Both expanded markedly over the 2000s, i.e., fishers from roughly 129,000 in 2000 to about 206,000 in 2014, and crafts from about 42,500 to 70,000, though the expansion was concentrated in the first half of the period, with effort broadly flat since around 2006. Importantly, this labour force is heavily concentrated in the most valuable species: in 2014, about 54% of fishers targeted Nile perch and a further 29% targeted dagaa, while tilapia accounted for roughly 13% and haplochromines for under 3%.

Panel (B) decomposes the fleet by means of propulsion. The number of engine-powered craft rose more than fivefold, from about 4,200 in 2000 to 21,600 in 2014, so that the motorized share climbed from roughly one in ten craft to nearly a third. Sails account for a small share throughout, with little net change. Despite the rise in motorization, however, the fishery remains predominantly human-powered: close to three-fifths of all craft were still paddled in 2014, and engine ownership is far from universal. Moreover, Panel (C) shows that the fishing capital is sharply sorted across species. Drawing on the Ugandan Frame Survey Reports, which tabulate propulsion by target species, motorization is overwhelmingly a Nile-perch phenomenon: the motorized share of Nile-perch craft rose from about 36% in 2006 to roughly 52% by 2012, against essentially none of those targeting tilapia. Another notable shift is the dagaa fishery, where the motorized share rose from under 3% in 2006 to about 31% in 2012.

Panel (D) indicates the productivity effects of choosing a more capital intensive technology, which is crucial for Nile perch catch. Catch per unit effort is substantially higher for motorized than for paddled vessels, and higher still when the engine is paired with longlines rather than gillnets. More capital-intensive technology thus raises catch productivity, and does so precisely for Nile perch, which is taken in deeper offshore waters and therefore requires fishers to travel further from shore.

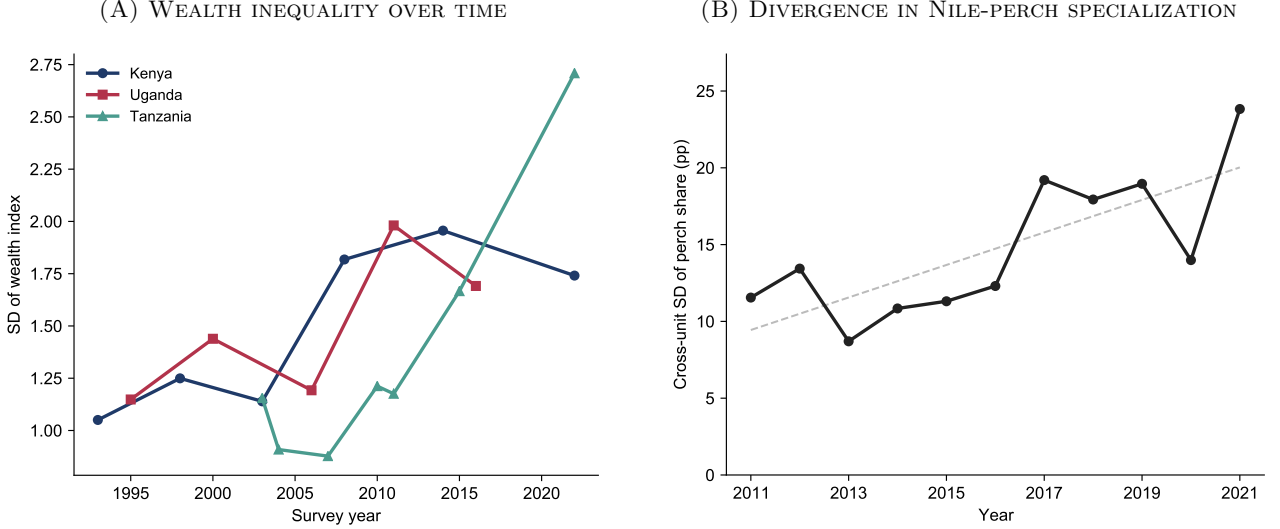
Taken together, these patterns describe a fishery economy in which the extensive margin, i.e., the number of boats and fishers, has plateaued since the mid-2000s, while the intensive margin, i.e., capital per vessel, has deepened, with that capital deployed disproportionately by those targeting the high-value predatory species. This heterogeneity in production technology across species, and the productivity gains it confers, motivates a model where fishing technologies differ in their level of capital intensity, and fishers choose endogenously which species to harvest.

Inequality and divergent fishing choice Our final set of facts examines the dynamics of wealth inequality among fishing households and the evolution of their species-targeting choices. Two regularities, summarized in Figure 4, are central.

Panel (A) documents the evolution of wealth inequality among fishing households along Lake Victoria. Using Demographic and Health Survey data for the lakeshore regions of Kenya, Uganda, and Tanzania, we summarize each fishing household’s wealth status and living standards by the first principal component of a standard set of assets and housing characteristics, and measure inequality as the standard deviation of this wealth index across fishing households within each survey wave. In all three countries, wealth index dispersion rises over the sample period, suggesting growing wealth inequality among fishing households capturing the common pool resources. It is useful to note that because the index is estimated separately by country, its level is not comparable across countries; but the common feature is the *within-country* upward trend.

Panel (B) turns to fishing behavior and documents a growing divergence in what fishers target. Using county-level fishing statistics for the five Kenyan Lake Victoria counties, we compute the share of Nile perch in each county’s total catch volume, and plot the cross-county standard deviation of this share over time. This dispersion roughly doubles between 2011 and 2021: counties that were broadly similar in their reliance on Nile perch increasingly diverge. Crucially, this divergence unfolds against a backdrop of falling aggregate Nile perch landings. As the lake-wide resource contracts, counties do not adjust in parallel; instead, some intensify their specialization in Nile perch while others diversify toward lower-value species such as dagaa (*Rastrineobola argentea*). Appendix Figure 13 shows this county by county: Migori moves further into Nile perch over the decade, whereas Homa Bay shifts sharply away from it toward dagaa, even though both face the same lake-wide decline. In fact, a

FIGURE 4: WEALTH INEQUALITY AND SPECIES SPECIALIZATION



Notes: Panel (A) is based on Demographic and Health Survey (DHS) data on fishing households in the Lake Victoria regions of Kenya (1993–2022), Uganda (1995–2016), and Tanzania (2003–2022). For each country we construct a household wealth index as the first principal component (pooled across that country’s waves, correlation matrix) of household assets (electricity, radio, television, refrigerator, bicycle) and housing quality; the plotted series is the standard deviation of this index across households in each survey wave. The indices are estimated separately by country and are not comparable in level across countries. Panel (B) is based on Kenya Fisheries Annual Statistical Bulletins, 2011–2021. For each county-year we compute Nile perch as a share of total catch volume (kg); the series is the cross-county standard deviation of that share (percentage points), with the dashed line a linear fit.

further decomposition of county-level Nile perch shares makes the pattern more precise: common year effects, i.e., the lake-wide aggregate component, explain only about 11% of the variation, while persistent cross-county differences account for roughly 54%, so specialization is largely a stable county characteristic rather than a shared response to the aggregate decline. Allowing each county its own linear trend raises the explained variation further to about 83%, and these county-specific trends differ sharply: Migori moves *into* Nile perch (about +2.8 percentage points per year) while Homa Bay moves *out* of it (about −3.3), with the hypothesis of equal trends strongly rejected (F -test, $p < 0.001$).

Taken together, these facts describe a setting in which wealth inequality among fishing households is rising while otherwise comparable communities diverge in their use of a shared, depleting resource. They motivate a theory in which household wealth, fishing and technology choices, and the dynamics of the fish stock evolve jointly: heterogeneity in wealth shapes which species and technologies households can profitably pursue, those choices feed back into the predator–prey dynamics of the lake, and the resulting ecological change in turn reshapes the returns to fishing and the distribution of wealth. The remainder of the paper develops and analyzes such a model.

3 The Model

3.1 Basics

We consider two groups of economic agents $i \in \{R, P\}$, namely, the rich and the poor. They differ only in their initial level of wealth/productive capital with $a_{R0} > a_{P0}$, and they take equal share of the population. Two species $s \in \{N, H\}$ are available for harvest, with μ_s denoting the stock of species s . Each period, agents inelastically supply one unit of time allocated between harvesting the

two species. Agent i 's lifetime utility over consumption is

$$\int_0^\infty e^{-\rho t} u(c_{it}) dt = \int_0^\infty e^{-\rho t} \left[\frac{c_{it}^{1-\gamma} - 1}{1-\gamma} \right] dt, \quad (1)$$

where $\rho > 0$ is the discount rate and $\gamma > 0$ is the coefficient of relative risk aversion. Unlike [Weitzman \(1998\)](#), [Metrick and Weitzman \(1998\)](#) and [Quaas and Requate \(2013\)](#), our model abstracts from any intrinsic amenity value of species diversity, i.e., the charismatic megafauna effect. Instead, we study an economy in which heterogeneous agents derive income directly from harvesting multiple species subject to capital constraints. We show that the resulting biodiversity conservation problem carries both efficiency and equity consequences.

Harvesting both species requires labor and capital, but the two technologies differ in capital intensity. Let $h_{si} \geq 0$ denote the share of group i 's labor allocated to harvesting species $s \in \{H, N\}$, with $h_{Hi} + h_{Ni} = 1$. Let $k_{si} \geq 0$ denote group i 's capital input in harvesting species s . Species H is relatively labor-intensive and can be harvested at small scale. The catch of species H is

$$q_i = q(k_{Hi}, h_{Hi}; \mu_H) = \mu_H^{1-\eta} \left(k_{Hi}^{\alpha_1} h_{Hi}^{1-\alpha_1} \right)^\eta, \quad (2)$$

where $\eta \in (0, 1]$ and $\alpha_1 \in (0, 1)$. Species N , by contrast, is more capital-intensive and requires a minimum operational scale. The catch of species N is

$$Q_i = \begin{cases} \mu_N^{1-\eta} \left((k_{Ni} - \underline{k})^{\alpha_2} h_{Ni}^{1-\alpha_2} \right)^\eta & \text{if } k_{Ni} \geq \underline{k}, \\ 0 & \text{if } k_{Ni} < \underline{k}, \end{cases} \quad (3)$$

where $\underline{k} > 0$ is the minimum capital requirement. We assume

$$0 < \alpha_1 < \alpha_2 < 1, \quad (4)$$

so that species N is more capital-intensive than species H . The parameter $\underline{k}$ captures the fixed scale required to operate the capital-intensive fishery, such as larger boats, specialized gear, fuel, and storage. The use of effective capital $k_{Ni} - \underline{k}$ ensures that catch from species N is continuous at the entry threshold. The minimum capital requirement introduces a non-convexity in the harvesting technology, giving rise to a poverty trap in the spirit of [Skiba \(1978\)](#) and [Dechert and Nishimura \(1983\)](#): agents with insufficient wealth to meet $\underline{k}$ are excluded from harvesting species N and may never accumulate enough capital to do so, remaining permanently in the low-wealth regime. Furthermore, the dependence of harvest on species stock is standard in the economics of fisheries literature ([Berck, 1979](#); [Karpoff, 1987](#); [Zhang and Smith, 2011](#); [Smith, 2012](#)).

Agents sell their harvest on international markets at exogenous, time-invariant prices p_N and p_H for species N and H , respectively. In the benchmark model, agents are self-financed: accumulated wealth can be used as productive capital, but agents cannot borrow or lend in an external capital market. Hence productive capital must satisfy $k_{Hi} + k_{Ni} \leq a_i$. Agent i 's budget constraint is

$$\dot{a}_i = p_N Q_i + p_H q_i - \delta a_i - c_i. \quad (5)$$

where $\delta > 0$ denotes the depreciation or operating cost of productive capital, and Q_i and q_i are defined in (3) and (2). The terms on the right-hand side represent, respectively, revenues from harvesting

species N and H , depreciation of capital, and consumption expenditure.

The stock dynamics of species μ_N and μ_H follow a classical predator-prey framework (Brown et al., 2005b; Quaas and Requate, 2013). The predator species N evolves according to

$$\dot{\mu}_N = \nu_1 \mu_N \left(1 - \frac{\mu_N}{\bar{N}}\right) + \theta_1 \mu_N \mu_H - Q_R - Q_P, \quad (6)$$

where $\bar{N}$ captures the natural carrying capacity of species N ⁶, and $\theta_1 > 0$ captures the fact that a larger prey stock raises predator growth. The prey species H evolves according to

$$\dot{\mu}_H = \nu_2 \mu_H \left(1 - \frac{\mu_H}{\bar{H}}\right) - \theta_2 \mu_H \mu_N - q_R - q_P, \quad (7)$$

where $\bar{H}$ is the carrying capacity of species H , and $\theta_2 > 0$ captures the fact that a larger predator stock depresses prey growth.

We proceed in two steps: we first characterize the laissez-faire equilibrium, then derive the social optimum and discuss how the former deviates from the latter, as well as the resulting efficiency and equity implications.

3.2 The Laissez-Faire Equilibrium

In the laissez-faire equilibrium, agents fail to internalize two distinct externalities. First, individual harvesting decisions do not account for the effect of current catches on future species stocks, giving rise to the classic open-access overfishing problem. Second, group i 's harvesting decision for species s imposes an ecological externality on group $-i$ through the predator-prey dynamics: by depleting one species, each group indirectly affects the stock, and thus the harvesting productivity, of the other species for the opposing group (Brown et al., 2005b).

The laissez-faire problem separates naturally into two components: a static, within-period harvesting problem in which agents choose k_{si} and h_{si} to maximize per-period profit given current species stocks, and a dynamic saving problem in which agents accumulate wealth taking per-period maximized profits as given. Crucially, agents treat μ_N and μ_H as fixed when making harvesting decisions, and therefore do not internalize the impact of current catches on future species stocks.

Static Harvesting Problem In each period, taking species stocks (μ_N, μ_H) and accumulated productive wealth a_i as given, group i allocates its capital and labor across the two harvesting activities. Let k_{si} and h_{si} denote the capital and labor allocated to species $s \in \{H, N\}$. The allocation constraints are

$$k_{Hi} + k_{Ni} \leq a_i, \quad h_{Hi} + h_{Ni} = 1, \quad (8)$$

with $k_{si} \geq 0$ and $h_{si} \geq 0$.

The gross harvesting income generated by wealth a_i is

$$Y(a_i; \mu_N, \mu_H) = \max_{\{k_{Hi}, k_{Ni}, h_{Hi}, h_{Ni}\}} \{p_H q(k_{Hi}, h_{Hi}; \mu_H) + p_N Q(k_{Ni}, h_{Ni}; \mu_N)\}, \quad (9)$$

subject to (8) and the minimum-scale requirement for species N .

Since both harvesting technologies are increasing in capital whenever they are operated, and since species H can be harvested at arbitrarily small scale, the capital constraint in (8) binds in any

⁶ $\bar{N}$ is the predator carrying capacity in the absence of prey-driven growth; because prey abundance raises predator growth, the steady-state predator stock may exceed $\bar{N}$.

optimum⁷. Now it is useful to characterize the static problem by considering two candidate regimes. First, if the group does not operate the capital-intensive technology for species N , then it allocates all capital and labor to species H :

$$k_{Hi} = a_i, \quad h_{Hi} = 1, \quad k_{Ni} = h_{Ni} = 0.$$

The corresponding gross income is simply

$$Y_H(a_i; \mu_H) = p_H q(a_i, 1; \mu_H) = p_H \mu_H^{1-\eta} a_i^{\eta \alpha_1}. \quad (10)$$

This regime is the only feasible harvesting regime for species N when $a_i < \underline{k}$.

Second, if the group operates species N , it must allocate at least $\underline{k}$ units of capital to that activity. Let $\tilde{k}_{Ni} \equiv k_{Ni} - \underline{k}$ denote effective capital in species N . Conditional on operating species N , the group solves

$$Y_M(a_i; \mu_N, \mu_H) = \max_{\tilde{k}_{Ni}, k_{Hi}, h_{Ni}, h_{Hi}} \left\{ p_H \mu_H^{1-\eta} \left(k_{Hi}^{\alpha_1} h_{Hi}^{1-\alpha_1} \right)^\eta + p_N \mu_N^{1-\eta} \left(\tilde{k}_{Ni}^{\alpha_2} h_{Ni}^{1-\alpha_2} \right)^\eta \right\} \quad (11)$$

subject to

$$k_{Hi} + \tilde{k}_{Ni} = a_i - \underline{k}, \quad h_{Hi} + h_{Ni} = 1, \quad (12)$$

with $k_{Hi}, \tilde{k}_{Ni}, h_{Hi}, h_{Ni} \geq 0$. An interior mixed allocation satisfies

$$p_H q_k(k_{Hi}, h_{Hi}; \mu_H) = p_N Q_k(k_{Ni}, h_{Ni}; \mu_N), \quad (13)$$

$$p_H q_h(k_{Hi}, h_{Hi}; \mu_H) = p_N Q_h(k_{Ni}, h_{Ni}; \mu_N), \quad (14)$$

where Q_k is the derivative with respect to effective capital $\tilde{k}_{Ni} = k_{Ni} - \underline{k}$. These conditions equate marginal revenue products of capital and labor across the two harvesting activities. Using the Cobb-Douglas forms, we have

$$\frac{\tilde{k}_{Ni}}{h_{Ni}} = \frac{\alpha_2(1-\alpha_1)}{\alpha_1(1-\alpha_2)} \frac{k_{Hi}}{h_{Hi}}. \quad (15)$$

Since $\alpha_2 > \alpha_1$, the coefficient on the right-hand side is greater than one. Thus, whenever both species are harvested, species N uses a higher capital-labor ratio than species H . Furthermore, we can show:

Lemma 1 *There exists a unique threshold $a^\dagger(\mu_N, \mu_H) > \underline{k}$ such that agents will harvest both species if and only if $a_i \geq a^\dagger(\mu_N, \mu_H) > \underline{k}$.*

The proofs are all in the Appendix. The indirect gross income function is therefore

$$Y(a_i; \mu_N, \mu_H) = \begin{cases} Y_H(a_i; \mu_H), & a_i < a^\dagger(\mu_N, \mu_H), \\ Y_M(a_i; \mu_N, \mu_H), & a_i \geq a^\dagger(\mu_N, \mu_H). \end{cases} \quad (16)$$

The minimum-scale requirement introduces a non-concavity into the indirect income function. Groups with low wealth remain specialized in the low-capital species H , while groups with wealth above the endogenous threshold $a^\dagger(\mu_N, \mu_H)$ operate the capital-intensive species N alongside species H . The

⁷To see this, suppose instead that $k_{Hi} + k_{Ni} < a_i$. If species H is harvested, reallocating a small amount of unused capital to H strictly raises $q(k_{Hi}, h_{Hi}; \mu_H)$. If species H is not harvested, then all labor is allocated to species N ; whenever species N is harvested, reallocating unused capital to N strictly raises $Q(k_{Ni}, h_{Ni}; \mu_N)$. Hence leaving productive wealth idle cannot be optimal. Therefore $k_{Hi} + k_{Ni} = a_i$.

threshold reflects both technological feasibility and profitability: even agents with $a_i \geq \underline{k}$ may prefer to remain in the H sector until wealth is sufficiently high to make the fixed scale cost worthwhile.

Dynamic Saving Problem Taking species stocks (μ_N, μ_H) as given, each group chooses consumption and accumulates productive wealth. The static harvesting problem characterized above delivers the indirect gross income function $Y(a_i; \mu_N, \mu_H)$. Accumulated productive wealth depreciates at rate δ , so the law of motion for wealth is

$$\dot{a}_i = Y(a_i; \mu_N, \mu_H) - c_i - \delta a_i. \quad (17)$$

The agent's dynamic problem is therefore to maximize (1) subject to (17).

Given species stocks (μ_N, μ_H) , the associated HJB equation is

$$\rho v(a; \mu_N, \mu_H) = \max_{c \geq 0} \{u(c) + v_a(a; \mu_N, \mu_H) [Y(a; \mu_N, \mu_H) - c - \delta a]\}. \quad (18)$$

At points where the value function is differentiable, the optimal consumption policy satisfies

$$u'(c^*) = v_a(a; \mu_N, \mu_H). \quad (19)$$

In the quantitative analysis, we solve the dynamic problem numerically⁸. When v is differentiable and the optimal policy is interior, the Euler equation is

$$\frac{\dot{c}_i}{c_i} = \frac{1}{\gamma} [Y_a(a_i; \mu_N, \mu_H) - \delta - \rho]. \quad (20)$$

Steady State It then follows that in any interior steady state with $\dot{c}_i = \dot{a}_i = \dot{\mu}_N = \dot{\mu}_H = 0$, we have

$$Y_a(a_i^*; \mu_N^*, \mu_H^*) = \rho + \delta, \quad (21)$$

$$Y(a_i^*; \mu_N^*, \mu_H^*) - \delta a_i^* = c_i^*, \quad (22)$$

$$\nu_1 \mu_N^* \left(1 - \frac{\mu_N^*}{N}\right) + \theta_1 \mu_N^* \mu_H^* = Q_R^* + Q_P^*, \quad (23)$$

$$\nu_2 \mu_H^* \left(1 - \frac{\mu_H^*}{H}\right) - \theta_2 \mu_H^* \mu_N^* = q_R^* + q_P^*. \quad (24)$$

We can characterize three types of steady states for the laissez-faire equilibrium. First, both groups are able to harvest species N :

$$a_R^* \geq a^\dagger(\mu_N^*, \mu_H^*), \quad a_P^* \geq a^\dagger(\mu_N^*, \mu_H^*). \quad (25)$$

The poor's wealth level will converge to the rich's, i.e., $a_R^* = a_P^* \equiv a_A^*$, and the steady state wealth is determined by

$$Y_{M,a}(a_A^*; \mu_N^*, \mu_H^*) = \rho + \delta, \quad (26)$$

where, by envelope theorem, $Y_{M,a}(a_A^*; \mu_N^*, \mu_H^*)$ is just the common marginal revenue product of capital

⁸The minimum-scale requirement for species N makes the indirect income function piecewise smooth, with a potential kink at the endogenous entry threshold $a^\dagger(\mu_N, \mu_H)$. Consequently, the value function may fail to be differentiable at this threshold. We therefore interpret (18) in the viscosity sense, with (19) applying at points of differentiability.

across both harvesting activities. The species-stock conditions become

$$\nu_1 \mu_N^* \left(1 - \frac{\mu_N^*}{N}\right) + \theta_1 \mu_N^* \mu_H^* = 2Q_M(a_A^*; \mu_N^*, \mu_H^*), \quad (27)$$

$$\nu_2 \mu_H^* \left(1 - \frac{\mu_H^*}{H}\right) - \theta_2 \mu_H^* \mu_N^* = 2q_M(a_A^*; \mu_N^*, \mu_H^*). \quad (28)$$

The more interesting case is where the rich harvest both species while the poor remain specialized in species H , i.e., an ecological inequality trap:

$$a_P^* < a^\dagger(\mu_N^*, \mu_H^*) \leq a_R^*. \quad (29)$$

We call this an ecological inequality trap because the ecological state and the wealth distribution jointly sustain asymmetric access to the capital-intensive fishery. The poor's steady state wealth is defined by

$$Y_{H,a}(a_P^*; \mu_H^*) = p_H(\mu_H^*)^{1-\eta} \eta \alpha_1 (a_P^*)^{\eta \alpha_1 - 1} = \rho + \delta, \quad (30)$$

which gives

$$a_P^* = \left[\frac{p_H(\mu_H^*)^{1-\eta} \eta \alpha_1}{\rho + \delta} \right]^{\frac{1}{1-\eta \alpha_1}}. \quad (31)$$

The rich's steady state wealth level is again characterized by (26). The steady state species stock are

$$\nu_1 \mu_N^* \left(1 - \frac{\mu_N^*}{N}\right) + \theta_1 \mu_N^* \mu_H^* = Q_M(a_R^*; \mu_N^*, \mu_H^*), \quad (32)$$

$$\nu_2 \mu_H^* \left(1 - \frac{\mu_H^*}{H}\right) - \theta_2 \mu_H^* \mu_N^* = q_M(a_R^*; \mu_N^*, \mu_H^*) + q_H(a_P^*; \mu_H^*). \quad (33)$$

Finally, consider the case in which both groups remain specialized in species H . This low-wealth steady state satisfies

$$a_R^* < a^\dagger(\mu_N^*, \mu_H^*), \quad a_P^* < a^\dagger(\mu_N^*, \mu_H^*). \quad (34)$$

Since both groups face the same H -only technology, their steady-state wealth levels coincide $a_R^* = a_P^* \equiv a_H^*$. The steady-state wealth level is again characterized by $Y_{H,a}(a_H^*; \mu_H^*) = p_H(\mu_H^*)^{1-\eta} \eta \alpha_1 (a_H^*)^{\eta \alpha_1 - 1} = \rho + \delta$, which implies $a_H^* = \left[\frac{p_H(\mu_H^*)^{1-\eta} \eta \alpha_1}{\rho + \delta} \right]^{\frac{1}{1-\eta \alpha_1}}$. Since neither group operates the capital-intensive fishery, the species-stock conditions now become

$$\nu_1 \mu_N^* \left(1 - \frac{\mu_N^*}{N}\right) + \theta_1 \mu_N^* \mu_H^* = 0, \quad (35)$$

$$\nu_2 \mu_H^* \left(1 - \frac{\mu_H^*}{H}\right) - \theta_2 \mu_H^* \mu_N^* = 2q_H(a_H^*; \mu_H^*). \quad (36)$$

For an interior steady state with $\mu_N^* > 0$, equation (35) can be written as

$$\mu_N^* = \bar{N} \left(1 + \frac{\theta_1}{\nu_1} \mu_H^*\right). \quad (37)$$

Here, in the absence of harvesting of species N , the predator stock may exceed its prey-free carrying capacity $\bar{N}$, because the prey stock raises predator growth. This steady state captures a low-wealth trapping outcome. Both groups are too poor to operate the capital-intensive technology profitably, so the economy remains specialized in the labor-intensive species H . In contrast to the previous two

types of steady state, inequality disappears in the long run, but this equality is achieved through common exclusion from the high-return species rather than through convergence into the capital intensive fishery.

We now summarize our definition of the dynamic equilibrium and characterization of the steady states:

Definition 1 (Dynamic laissez-faire equilibrium and steady states) *Given initial conditions $(a_{R0}, a_{P0}, \mu_{N0}, \mu_{H0})$, a dynamic equilibrium is a collection of paths $\{a_{it}, c_{it}, k_{Hit}, k_{Nit}, h_{Hit}, h_{Nit}, q_{it}, Q_{it}\}_{t \geq 0}$ and $\{\mu_{Nt}, \mu_{Ht}\}_{t \geq 0}$ such that for each $i \in \{R, P\}$:*

- (i) *Given $(a_{it}, \mu_{Nt}, \mu_{Ht})$, harvesting choices solve the static problem (9).*
- (ii) *Given $Y(a_{it}; \mu_{Nt}, \mu_{Ht})$, consumption and wealth accumulation solve the dynamic saving problem (18) subject to (17).*
- (iii) *Species stocks evolve according to (6) and (7).*
- (iv) *The feasibility constraints in (8) hold, together with non-negativity of consumption, capital inputs, labor inputs, catches, wealth, and species stocks.*

A steady state is a time-invariant dynamic laissez-faire equilibrium with $\dot{a}_i = \dot{c}_i = \dot{\mu}_N = \dot{\mu}_H = 0$ for $i \in \{R, P\}$. At any interior steady state, the general steady-state restrictions are given by (21)–(24). We further distinguish three steady-state regimes.

- (i) *In an egalitarian steady state, both groups operate the capital-intensive fishery with condition (25) and $a_R^* = a_P^* \equiv a_A^*$. $(a_A^*, \mu_N^*, \mu_H^*)$ is characterized by (26)–(28), and consumption is determined by (22).*
- (ii) *In an ecological inequality trap with the regime condition (29), the poor group remains specialized in species H , while the rich group operates the mixed technology. The poor group's wealth is characterized by (30), equivalently by (31). The rich group's wealth is characterized by the rich-group analogue of (26), i.e., $Y_{M,a}(a_R^*; \mu_N^*, \mu_H^*) = \rho + \delta$. The species stocks are characterized by (32)–(33), and consumption for both groups is determined by (22).*
- (iii) *In a subsistence steady state, both groups remain specialized in species H , satisfying the regime condition (34). Since both groups face the same H -only technology, their wealth levels coincide, $a_R^* = a_P^* \equiv a_H^*$. The common wealth level is characterized by $a_H^* = \left[\frac{p_H(\mu_H^*)^{1-\eta}\eta\alpha_1}{\rho+\delta} \right]^{\frac{1}{1-\eta\alpha_1}}$. The species stocks are characterized by (35)–(36). Consumption for both groups is determined by (22), with $Y(a_H^*; \mu_N^*, \mu_H^*) = Y_H(a_H^*; \mu_H^*)$.*

3.3 The Constrained Social Planner's Problem

We now formulate the social planner's problem and characterize the constrained-efficient allocation. The planner faces the same harvesting technologies, minimum-scale requirement, ecological laws of motion, and group-specific capital accumulation constraints as in the laissez-faire economy. The difference is that the planner chooses the harvesting regimes and inputs of both groups to maximize aggregate utility, internalizing the effect of current harvesting on future species stocks.

We denote the state variables as $x = (a_R, a_P, \mu_N, \mu_H)$. Let $r_i \in \{H, M\}$, where H denotes specialization in species H , and M denotes harvesting both species, i.e., the mixed regime. A harvest regime profile is $r = (r_R, r_P) \in \{H, M\}^2$. Furthermore, at state x , a regime profile r is feasible

only if every group assigned to the mixed regime has enough wealth to meet the minimum capital requirement:

$$\mathcal{R}(x) = \{r \in \{H, M\}^2 : r_i = M \Rightarrow a_i \geq \underline{k} \text{ for each } i \in \{R, P\}\}. \quad (38)$$

Thus, for example, if $a_P < \underline{k}$, any regime profile with $r_P = M$ is infeasible.

The social planner's problem can be formulated as follows: given an initial state $x_0 = (a_{R0}, a_{P0}, \mu_{N0}, \mu_{H0})$, the planner chooses paths

$$\{r_t, c_{it}, k_{Hit}, k_{Nit}, h_{Hit}, h_{Nit}\}_{t \geq 0, i \in \{R, P\}, r_t \in \mathcal{R}(x_t)}$$

to maximize the weighted counterpart of (1),

$$\int_0^\infty e^{-\rho t} \sum_{i \in \{R, P\}} \omega_i u(c_{it}) dt,$$

subject to the group-specific capital accumulation constraint (5), the ecological laws of motion (6)–(7), the harvesting technologies (2)–(3), the allocation constraints (8)⁹, and non-negativity constraints. ω_i is the social weight for group i . It is important to note that the regime profile is a discrete control chosen by the planner at each state. Thus the optimal policy induces a state-dependent regime choice $r^*(x) \in \mathcal{R}(x)$, and the regime profile may change over time along the transition path as the state variables evolve. This formulation makes the comparison with laissez-faire economy transparent. In laissez faire, each group chooses whether to enter the capital-intensive fishery by comparing private returns, summarized by the threshold $a^\dagger(\mu_N, \mu_H)$ in (16). The planner instead compares the social value of the feasible regime profiles, taking into account how current catches affect the future evolution of μ_N and μ_H through (6) and (7), as well as their effect on the distribution of consumption across the two groups. Therefore the extensive margin of harvest choice is one of the key departures of laissez-faire equilibrium from the constrained social optimum.

The recursive representation of this sequence problem is given by the HJB equation below. Let $W(x)$ denote the planner's value function, then

$$\begin{aligned} \rho W(x) = \max_{r \in \mathcal{R}(x)} \max_{\{c_i, k_{Hi}, k_{Ni}, h_{Hi}, h_{Ni}\}_{i \in \{R, P\}}} & \left\{ \sum_{i \in \{R, P\}} \omega_i u(c_i) \right. \\ & + \sum_{i \in \{R, P\}} W_{a_i}(x) [p_N Q_i + p_H q_i - \delta a_i - c_i] \\ & + W_{\mu_N}(x) \left[\nu_1 \mu_N \left(1 - \frac{\mu_N}{N} \right) + \theta_1 \mu_N \mu_H - Q_R - Q_P \right] \\ & \left. + W_{\mu_H}(x) \left[\nu_2 \mu_H \left(1 - \frac{\mu_H}{H} \right) - \theta_2 \mu_H \mu_N - q_R - q_P \right] \right\}, \end{aligned} \quad (39)$$

where the inner maximization is subject to the feasibility constraints implied by the chosen regime profile r .

Another departure of the laissez-faire equilibrium from the social optimum is the intensive margin of harvest decision. We can see this from planner's first order conditions. Conditional on a feasible

⁹This implies that if $r_{it} = H$, then group i specializes in species H : $k_{Nit} = h_{Nit} = 0, h_{Hit} = 1, Q_{it} = 0$. If $r_{it} = M$, then group i operates the mixed technology and must satisfy the minimum-scale requirement in (3).

active regime profile $r^* \in \mathcal{R}(x)$, and assuming differentiability and interior continuous controls, the planner's consumption choice satisfies

$$\omega_i u'(c_i) = W_{a_i}(x),$$

where $W_{a_i}(x)$ is the social marginal value of productive wealth for group i . If group i is assigned to the mixed regime, $r_i^* = M$, and the allocation constraint (8) is binding¹⁰, its allocation of capital and labor across species satisfies

$$[p_H W_{a_i}(x) - W_{\mu_H}(x)] q_k(k_{Hi}, h_{Hi}; \mu_H) = [p_N W_{a_i}(x) - W_{\mu_N}(x)] Q_k(k_{Ni}, h_{Ni}; \mu_N), \quad (40)$$

$$[p_H W_{a_i}(x) - W_{\mu_H}(x)] q_h(k_{Hi}, h_{Hi}; \mu_H) = [p_N W_{a_i}(x) - W_{\mu_N}(x)] Q_h(k_{Ni}, h_{Ni}; \mu_N). \quad (41)$$

Here, the social marginal value of an additional unit of catch from each species is $\mathcal{P}_{Hi}(x) \equiv p_H W_{a_i}(x) - W_{\mu_H}(x)$ and $\mathcal{P}_{Ni}(x) \equiv p_N W_{a_i}(x) - W_{\mu_N}(x)$. An additional unit of catch from species s raises group i 's wealth by its market value p_s , but lowers the corresponding species stock by one unit. Hence the planner values catch at its market value net of the shadow cost of depleting the species stock.

These conditions are the planner's analogue of the laissez-faire static conditions (13)–(14). The difference is that private agents equate marginal revenue products evaluated at market prices, whereas the planner equates marginal products evaluated at social values. Thus the intensive-margin wedge is

$$\mathcal{P}_{si}(x) - p_s W_{a_i}(x) = -W_{\mu_s}(x), \quad s \in \{H, N\}.$$

If $W_{\mu_s}(x) > 0$, harvesting species s imposes a positive ecological shadow cost, so laissez faire overvalues current catch relative to the planner. Because the two species are linked through predator-prey dynamics, the shadow values $W_{\mu_N}(x)$ and $W_{\mu_H}(x)$ reflect not only the direct future productivity of each species, but also its effect on the other species.

To summarize, there are therefore two distinct deviations of laissez faire from the constrained-efficient allocation. First, conditional on a harvesting regime, private agents choose the intensive allocation of capital and labor using market prices rather than social shadow values. Second, because operating species N requires the fixed capital threshold $\underline{k}$, private agents may enter or remain excluded from the mixed regime at wealth levels that differ from those chosen by the planner. The first wedge affects how much pressure each active group places on the two species; the second wedge affects which groups harvest which species in the first place.

We can also characterize the steady state(s) in the social planner's problem:

Definition 2 (Social Planner's Steady State) *A planner steady state is a time-invariant solution to the planner's HJB (39). It consists of a state $x^* = (a_R^*, a_P^*, \mu_N^*, \mu_H^*)$, a feasible regime profile $r^* \in \mathcal{R}(x^*)$, and associated controls such that r^* and the continuous controls solve the maximization problem in (39), the wealth laws (5) and the ecological laws (6)–(7) are stationary, and all feasibility and regime-specific constraints are satisfied.*

¹⁰We argued that in the laissez faire economy, the economic agents will use all its productive capital into production, because both harvest technologies are increasing in capital and agents can obtain higher income from marginally higher harvest because market prices are positive. However, in the social planner's problem, the relevant "price" is the social marginal value of catch, hence it is possible that in some time periods, the social planner may not use all the productive wealth into harvest to protect future species stock, when such social marginal value of catch is negative. However we show in Appendix B.3 that as long as at least one species has strictly positive social marginal catch value, then the allocation constraints (8) must be binding. In our numerical exercises, we focus on the parameter space that such constraints are binding (to be more comparable to the laissez-faire economy).

The Implementation of Social Optimum We now discuss how one may implement the (constrained) socially optimal harvesting decisions in a decentralized, laissez-faire economy. For simplicity, we set equal welfare weights, $\omega_R = \omega_P = 1$. It then naturally follows that the implementation requires the species- and group-specific Pigouvian harvest taxes. For each species $s \in \{H, N\}$ and group $i \in \{R, P\}$, let

$$\tau_{si}(x) = \frac{W_{\mu_s}(x)}{W_{a_i}(x)}. \quad (42)$$

Then the effective price received by group i from harvesting species s is $\tilde{p}_{si}(x) = p_s - \tau_{si}(x)$. Under these taxes, private agents evaluate catches using the social marginal value of the corresponding species stock. Thus, conditional on a given regime profile, the decentralized first-order conditions coincide with the planner's first-order conditions. If $W_{\mu_s}(x) > 0$, harvesting species s reduces a valuable future stock and the optimal instrument is a positive harvest tax. If $W_{\mu_s}(x) < 0$, the optimal instrument is instead a subsidy. This may occur in the predator-prey system because a larger stock of one species can depress the future stock of the other. In such states, harvesting species s has a beneficial ecological effect, so the planner encourages rather than discourages current catch.

Furthermore, we also require that each agent's tax revenue rebated lump-sum to itself so that the after tax income and the associated wealth accumulation path under the laissez-faire coincides with the social optimum. However, it is important to note that the fact that the Pigouvian taxes can implement the social optimum relies on the assumption that $\bar{k}$ is a *feasibility* threshold rather than a sunk adoption cost, and the capital intensive technology is freely reversible. This implies that the regime choice r is a within-instant control, chosen point-wise to maximize the instantaneous Hamiltonian. Therefore the after tax optimality condition for technology choice under laissez-faire will coincide with that of the constrained social planner¹¹.

3.4 Alternative Policies

The first-best implementation described above decentralizes the planner's harvest decisions, but it requires strong assumptions on information and enforcement of the state. In particular, the government must observe the state $x = (a_R, a_P, \mu_N, \mu_H)$, compute the relevant shadow values $W_{a_i}(x)$ and $W_{\mu_s}(x)$, and implement instruments that vary across groups, species, and states. These requirements may be demanding in the institutional context of small-scale fisheries with limited state capacity.

We therefore consider simpler second-best policy instruments that require less targeting and less information about the wealth distribution. These are also policy proposals central to the ongoing policy debate in Lake Victoria (see, for example, [Brown et al. \(2005a\)](#) for a discussion of introducing fishing rights in Lake Victoria). However, these policies do not generally implement the constrained social optimum. Instead, they provide one-sided tools that affect the incentives to harvest particular species N . In the quantitative analysis, we vary the stringency of each policy and study how close it can reshape the decentralized economy, as well as its implications for efficiency, species conservation, and inequality.

¹¹If instead taking up the capital-intensive technology carries a fixed or sunk cost, or is irreversible, we would in general require an extensive-margin transfer $T_i^M(x)$ chosen so that the planner's desired regime is privately optimal: $Y_M^T(a_i; \mu_N, \mu_H) + T_i^M(x) \geq Y_H^T(a_i; \mu_H)$ if $r_i^*(x) = M$, and $Y_M^T(a_i; \mu_N, \mu_H) + T_i^M(x) \leq Y_H^T(a_i; \mu_H)$ if $r_i^*(x) = H$. Here, given the Pigouvian taxes above, $Y_H^T(a_i; \mu_H)$ and $Y_M^T(a_i; \mu_N, \mu_H)$ are the indirect gross incomes from the specialized and mixed regimes evaluated at the effective prices $\tilde{p}_{Hi}(x)$ and $\tilde{p}_{Ni}(x)$. Hence, when the planner wants a group to enter the mixed regime but private returns are too low, $T_i^M(x) > 0$ acts as an entry subsidy. Conversely, when private entry into the mixed regime generates excessive ecological costs, $T_i^M(x) < 0$ acts as an entry tax or license fee.

Minimum mesh size The first policy is a minimum mesh-size requirement. We model this policy as a uniform restriction that requires a fraction $\kappa \in [0, 1]$ of the catch of species N to be released. Equivalently, if group i catches Q_i , only $(1 - \kappa)Q_i$ can be landed and sold. Under this policy, group i 's budget constraint becomes

$$\dot{a}_i = p_N(1 - \kappa)Q_i + p_H q_i - \delta a_i - c_i.$$

The ecological law for species N is modified accordingly:

$$\dot{\mu}_N = \nu_1 \mu_N \left(1 - \frac{\mu_N}{N}\right) + \theta_1 \mu_N \mu_H - (1 - \kappa)(Q_R + Q_P).$$

The law of motion for species H remains given by (7). Thus the policy lowers both the private return to harvesting species N and the effective depletion of the predator stock.

Conditional on operating the capital intensive technology, the static harvesting problem is the same as in (11), except that the effective market price of species N is reduced from p_N to $(1 - \kappa)p_N$. Hence a higher κ discourages entry into the mixed regime and reduces harvesting pressure on species N . However, because N and H are linked through the predator-prey system, the welfare effect is ambiguous ex ante. Protecting species N may improve future harvest in the N -fishery, but it may also increase predation pressure on species H . Similarly, we can also introduce a mesh size requirement for the labor intensive species H , and we will see how such one-sided conservation policies have different aggregate and distributional implications.

Tradable fishing quota The second policy is a tradable quota for harvesting species N . Let $\sigma_0 \geq 0$ denote the quota initially allocated to each group. The total quota supply is therefore $2\sigma_0$. Quotas are perfectly tradable between the two groups at price p^σ . If group i harvests Q_i units of species N , it must hold Q_i units of quota. Its net quota purchase is $Q_i - \sigma_0$, so its wealth evolves according to

$$\dot{a}_i = (p_N - p^\sigma)Q_i + p_H q_i + p^\sigma \sigma_0 - \delta a_i - c_i. \quad (43)$$

Thus the quota price reduces the effective private return to harvesting species N , while the initial quota allocation generates a lump-sum transfer $p^\sigma \sigma_0$ to each group.

Given the quota price p^σ , group i 's static harvesting problem is therefore equivalent to the laissez-faire harvesting problem with the effective price of species N replaced by $p_N - p^\sigma$. The quota market clears according to

$$Q_R + Q_P = 2\sigma_0, \quad (44)$$

whenever the aggregate quota constraint binds. The ecological laws of motion remain (6)–(7), with $Q_R + Q_P$ determined by the quota-constrained decentralized equilibrium.

The tradable quota differs from the mesh-size policy in two ways. First, it directly controls aggregate harvesting of species N , while allowing the market to allocate harvesting rights across groups. Second, because quotas are initially allocated equally, the policy has a redistributive component. If the poor group does not operate the capital-intensive fishery, it can sell its quota to the rich group and receive quota income. Hence the quota system can relax the ecological externality while also partially transferring income toward the group excluded from species N .

In the quantitative analysis, we solve the decentralized equilibrium under different values of κ and σ_0 . We then compare the resulting transition paths and steady states with both the laissez-faire equilibrium and the constrained social optimum. This comparison allows us to evaluate how

TABLE 1: CALIBRATION

Parameter	Symbol	Value	Source / target
<i>Preferences and depreciation</i>			
Discount rate	ρ	0.04	standard annual value
Relative risk aversion	γ	1.5	standard CRRA value
Depreciation	δ	0.05	standard annual value
<i>Harvest technology</i>			
Returns to the effort aggregate	η	0.60	0.59-1.23 (Hannesson, 1983; Aljafary et al., 2019)
Capital share, prey tech.	α_1	0.55	labour-intensive $\alpha_1 < \alpha_2$
Capital share, perch tech.	α_2	0.70	capital-intensive $\alpha_1 < \alpha_2$
<i>Prices</i>			
Beach price, perch	p_N	0.10	normalization
Beach price, prey	p_H	0.04	relative price $p_N/p_H = 2.5$
<i>Bio-economic block</i>			
Intrinsic growth, perch	ν_1	0.18	slow predator (Brown et al., 2005b)
Intrinsic growth, prey	ν_2	1.10	fast prey (Brown et al., 2005b)
Carrying capacity, perch	$\bar{N}$	30	normalization
Carrying capacity, prey	$\bar{H}$	90	normalization
Predator gain	θ_1	0.006	predation-fed
Predation loss	θ_2	0.020	$\theta_2 \approx 3\theta_1$, as in Brown et al. (2005b)
Minimum capital, perch	$\bar{k}$	1.3	non-convex entry threshold; set to generate the two regimes
<i>Initial (pre-boom) ecology</i>			
Initial perch stock	μ_{N0}	4	low pre-boom predator stock
Initial prey stock	μ_{H0}	75	abundant pre-boom prey

much of the planner’s welfare gain can be achieved by simpler policy instruments, and whether these instruments improve or worsen the distributional consequences of ecological regulation.

3.5 Calibration

We now discuss how we calibrate the parameters. Preferences and capital depreciation (ρ, δ, γ) take standard values. The harvest technology $(\eta, \alpha_1, \alpha_2)$ is disciplined by the empirical fishery economics literature. The bioeconomic block is calibrated in the spirit of Brown et al. (2005b), whose predator–prey model is calibrated for the same fishery (Nile perch preying on the native haplochromine cichlids or dagaa), with the intrinsic growth and predation parameters adjusted so that the calibrated economy reproduces the two qualitative regimes of interest, i.e., an egalitarian steady state, and the ecological inequality trap. Table 1 collects the values.

Preferences and depreciation. The discount rate $\rho = 0.04$, depreciation $\delta = 0.05$, and risk aversion $\gamma = 1.5$ take conventional values.

Harvest technology. Each harvest technology is Cobb–Douglas in capital and labour, nested alongside a stock term, so the catch has a stock-output elasticity $1 - \eta$ and an effort-output elasticity η . We set $\eta = 0.6$, so that catch is concave in the composite harvesting input. This value is close to the lower end of empirical effort elasticities estimated in Cobb–Douglas fishery production functions (Aljafary et al., 2019; Eide et al., 2003; Hannesson, 1983). The key technological restriction is $\alpha_2 > \alpha_1$: the perch technology is more capital-intensive than the prey technology. But credible estimates for capital

and labor shares in the fishery production function is rare; we assume $\alpha_2 = 0.7$ and $\alpha_1 = 0.55$.

Prices. Nile perch is the high-value export species and the haplochromine prey the low-value species sold for local consumption. We normalize $p_N = 0.10$, and given our empirical facts on prices in Figure 2, we set $p_N/p_H = 2.5$. Therefore $p_H = 0.04$ for our benchmark model. However, in Section 4.5, we will introduce inverse demand functions for the two resources and allow p_{Nt} and p_{Ht} to be endogenously evolving.

Bio-economic block. The ecology is the predator–prey system of Brown et al. (2005b), calibrated for the same fishery. We retain its two defining features, i.e., a slow-growing predator and a fast-growing prey ($\nu_1 \ll \nu_2$), and a predation-loss term roughly three times the predator gain ($\theta_2 \approx 3\theta_1$), and adjust the levels so that the calibrated fishery is interior and sustainable at the model’s economic scale: $\nu_1 = 0.18$, $\nu_2 = 1.10$, $\theta_1 = 0.006$, $\theta_2 = 0.020$. With these values Nile perch stock is sustained primarily by predation rather than own logistic growth, consistent with the historical perch boom and the collapse of the haplochromine stock. The carrying capacities $\bar{N} = 30$, $\bar{H} = 90$ set the scale of the two stocks.

The non-convexity and the targeted trap. The minimum capital requirement $\bar{k} = 1.30$ is the central parameter: a group is able to the perch technology only if $a_i \geq \bar{k}$. We did not have a reliable market price for a motorized perch craft to pin $\bar{k}$ externally; we instead set it, together with the price scale, so that the decentralized laissez-faire economy exhibits all three regimes of interest across the relevant range of initial wealth. But in the extension exercises in Section 4.5, we allow $\bar{k} \in \{1.2, 1.4\}$ to vary and see how this changes steady state selection. The pre-boom initial ecology ($\mu_{N0} = 4$, $\mu_{H0} = 75$: scarce predator, abundant prey) implies the reported transitions trace the response to the historical Nile perch boom.

3.6 Solution Algorithm

Because the minimum-capital requirement $\bar{k}$ makes the problem non-convex, we solve the model, i.e., the HJB equation, with finite difference methods. For each regime, i.e., laissez-faire and the constrained social optimum, we compute the value function on a four-dimensional grid over the wealth levels and the fish stocks (a_R, a_P, μ_N, μ_H) by implicit-upwind finite-difference dynamic programming, in the spirit of Achdou et al. (2022). Within each iteration the agents’ consumption–saving and harvesting choices are taken from the first-order conditions evaluated at the same upwind marginal value of wealth, and the discrete regime choice, i.e., whether to operate the capital-intensive perch technology or specialize in the prey, is resolved as a pointwise maximum over the two branches, which is the numerical counterpart of the non-convexity. The difference between laissez-faire and social optimum is that the planner’s harvest internalizes the shadow values of the two stocks (the gradients W_{μ_N}, W_{μ_H} of its own value function), whereas these are set to zero under laissez-faire; the two solutions are therefore directly comparable, and we warm-start the planner from the converged LF value function to speed and stabilize convergence.

Given a converged value function, the optimal policies are smooth feedback rules on the grid: consumption from the marginal value of wealth, and the catches of each species from the harvesting first-order conditions. A transition path from any initial condition is then obtained by forward-integrating the controlled state system, i.e., wealth accumulation at market prices for each group together with the predator–prey stock dynamics, using these interpolated feedback rules, under either the laissez-faire or the social optimum policy. The long-run outcome to which a given initial condition converges is identified as the rest point of this forward simulation: we integrate to a long horizon and

read off the steady state the economy settles into, i.e., the egalitarian steady state or the ecological inequality trap¹², cross-checking against the stationarity conditions of the state system.

To map how the long-run outcome depends on the starting point, we fix the initial stocks at their pre-boom level (μ_{N0}, μ_{H0}) and discretize the space of initial wealth pairs (a_{R0}, a_{P0}) on a grid. Here, we require $a_{R0} > a_{P0}$ and additionally $a_{R0} \geq \bar{k}$, i.e., the rich group is able to take up the capital intensive technology for period 0. For each grid point we run the forward simulation described above and record the steady state it converges to, doing this separately under the laissez-faire and the social optimum policy. This produces, for each regime, a classification of initial conditions by their eventual outcome; overlaying the two classifications shows where laissez-faire and the planner agree on the destination and where they diverge.

With these in mind, we now present our quantitative results.

4 Quantitative Results

The results section characterizes the laissez-faire economy and the constrained social optimum, and present policy counterfactuals. We proceed in four steps. First, we compare the long-run ecology under the two regimes at each type of steady states. Second, we map which initial conditions lead to which steady state, under both laissez-faire and the planner, and show where the two agree and disagree as well as the welfare implications. Third, we unpack this further through the transition dynamics, and use them to describe how the constrained optimum can be implemented in a decentralized, laissez-faire economy through Pigouvian harvest taxes. Finally, we quantify the welfare and distributional consequences of alternative policy instruments available to a real regulator, i.e., a tradable harvest quota and a minimum-mesh-size regulation.

4.1 Steady-State Ecology

We begin with the long-run fish stocks, comparing the laissez-faire allocation with the constrained social optimum within each of the two interior steady states (Figure 5).

In the egalitarian steady state (Panel A), where both groups operate the capital-intensive Nile perch technology, the two steady state allocations are nearly indistinguishable: the stocks of both species under laissez-faire and the social optimum coincide to within roughly one to two percent. The reason the planner has so little to correct is that the shadow value of the predator stock is *positive* in this regime, i.e., the Nile perch is simply a valuable harvestable asset for both groups, and unlike the inequality trap below, there is no distributional motive to manage the predator/prey differently. For each species, the social-optimum-versus-laissez-faire difference combines a direct own-conservation effect with an indirect trophic effect through the other species, and in this regime the two work in *opposite* directions. For the predator, conserving it directly raises the perch stock, but the slightly thinner prey base (see below) means less food for perch, which pulls it back down; the two nearly cancel and μ_N is essentially unchanged across the two regimes. For the prey, the planner conserves it directly, but by also conserving the predator it raises predation pressure, which works the opposite way; here the predation channel marginally dominates, so the prey stock comes out, if anything, slightly *higher* under laissez-faire, not because laissez-faire conserves the prey, but because its more aggressive harvesting of the predator leaves weaker predation and thus a little more prey as a byproduct.

The inequality trap (Panel B) is different: there the social optimum sustains *higher* stocks of both

¹²Given all our initial conditions, we didn't find the subsistence steady state.

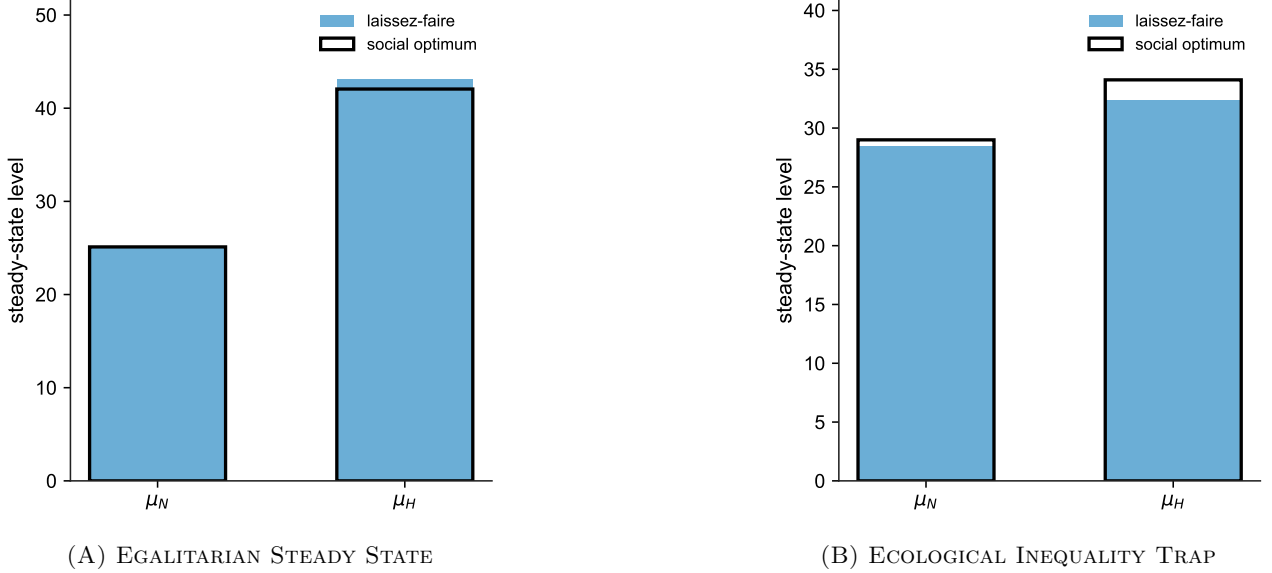


FIGURE 5: STEADY-STATE FISH STOCKS

Notes: Each panel reports the long-run stocks of the two species, i.e., the predator μ_N (Nile perch) and its prey μ_H (haplochromine cichlids), at the steady state to which the economy converges, comparing the laissez-faire allocation with the constrained social optimum. In each panel, the solid blue bars denote the laissez-faire allocation and the black-outlined bars the social optimum. Panel (A) shows the egalitarian regime, in which both groups operate the capital-intensive perch technology; Panel (B) shows the “ecological inequality trap,” in which the wealthy operate the perch fishery while the asset-poor are confined to the prey fishery. Stocks are computed by solving each agent’s HJB problem on a discretized four-dimensional state space (a_R, a_P, μ_N, μ_H) and forward-integrating the resulting optimal feedback rules from the relevant initial condition until the state system reaches its rest point.

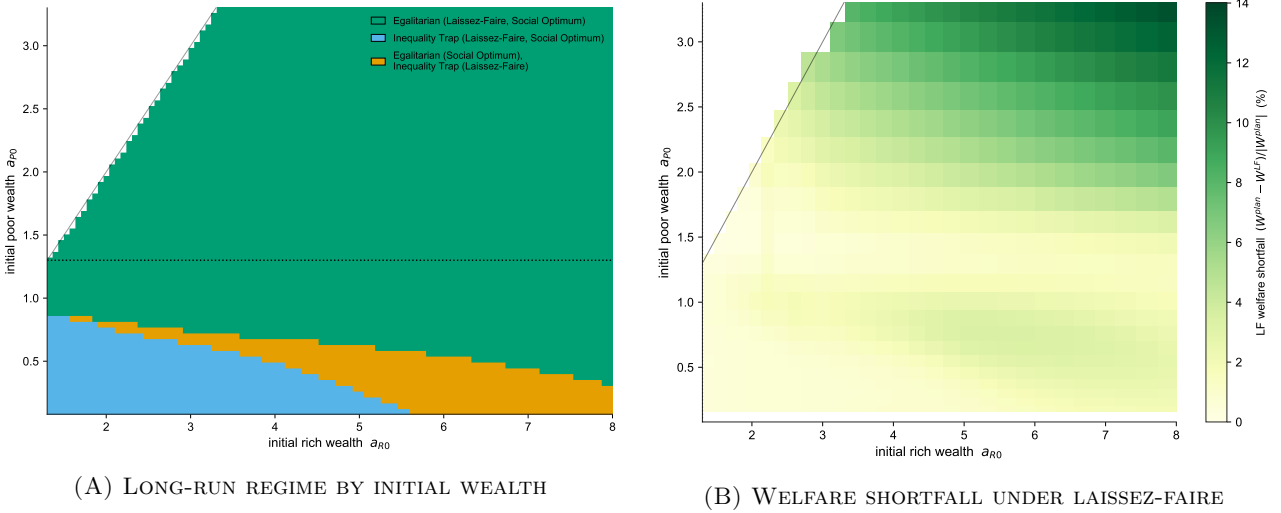
species than laissez-faire. Two forces work in the same direction here. The planner conserves the prey on which the poor specializes, and, unlike in the egalitarian regime, it now also harvests the predator down rather than conserving it, because a large Nile perch stock suppresses the haplochromine prey and hence sustains the trap. Culling the predator relieves predation pressure on the prey, which reinforces the direct conservation of the prey itself; the prey stock therefore rises markedly under the social optimum. The predator stock is also slightly higher under the planner, since the larger prey base feeds perch growth and outweighs the planner’s direct harvesting of it.

The contrast between the two panels traces back to the catch technology choices and the implied social marginal value of species stock. By determining which groups fish the predator and which are confined to the prey, the social planner incorporates the distributional role of the Nile perch which then determines the shadow value of species. Such social value is positive in the egalitarian regime, where the perch is a shared asset, and negative in the inequality trap, where a larger predator stock suppresses the prey on which the poor depend. This then propagates through the predator-prey system to shape the long-run ecology.

4.2 Initial Wealth Distribution and Steady-State

The two steady states differ in their pattern of species catch choice, and which catch technology a group adopts is governed by its wealth level (e.g., relative to the minimum-capital threshold $\bar{k}$). It is therefore natural to ask, across the entire space of initial wealth, which steady state the economy selects. We find that each initial condition is associated with a *unique* steady state, reached along a determinate transition path. In addition, we can also verify that these steady states are at least locally saddle-path stable. Panel (A) of Figure 6 maps the steady state selection, under both laissez-faire

FIGURE 6: STEADY-STATE SELECTION AND THE WELFARE COST OF LAISSEZ-FAIRE



Notes: Both panels vary the initial wealth pair (a_{R0}, a_{P0}) , holding the pre-boom fish stocks (μ_{N0}, μ_{H0}) fixed. Panel (A) reports the long-run regime the economy converges to from each initial condition, obtained by forward-integrating the optimal feedback rules to their rest point and classifying the resulting steady state. Green denotes initial conditions from which both laissez-faire and the social optimum reach the egalitarian steady state; blue, those from which both reach the inequality trap; and orange, those from which the social optimum reaches the egalitarian steady state while laissez-faire is absorbed into the trap. The dotted line marks the minimum-capital threshold $\bar{k}$. The regimes are computed by solving the agents' HJB problem on a discretized four-dimensional state space (a_R, a_P, μ_N, μ_H) and classifying the resulting long-run outcomes under laissez-faire and the constrained social optimum. Panel (B) reports the welfare loss of laissez-faire relative to the social optimum, $(W^{plan} - W^{LF})/|W^{plan}|$, in percent, where W is the discounted lifetime aggregate welfare, i.e., the present value of the two groups' flow utilities at equal weights, evaluated along the optimal path from each initial condition. Darker shading indicates a larger welfare loss.

and the social optimum, over the plane of initial wealth.

When the poor begin sufficiently wealthy, the economy converges to the egalitarian steady state under both regimes (green). This includes initial conditions at which the poor start *below* the threshold $\bar{k}$ and cannot initially operate the perch fishery: by harvesting the prey alone they accumulate wealth, climb across the feasibility threshold during the Nile perch boom, take up the capital-intensive technology, and converge to the rich. Where the poor are wealthy enough to make that climb, laissez-faire and the planner also agree.

On the other hand, when the poor begin much poorer, laissez-faire instead delivers the inequality trap (blue and orange). Starting with too little wealth, the poor cannot accumulate quickly enough during the boom to reach $\bar{k}$; later on the window closes as Nile perch stock declines, and they remain confined to the prey fishery indefinitely while the rich operate the perch technology. In the blue region, the social planner would also agree with the laissez-faire economy on the inequality-trap steady state. The constrained planner cannot simply transfer wealth to lift the poor across $\bar{k}$; its only levers are the harvest allocations across the two groups and, through them, the trajectory of the fish stocks. Raising the poor's prey income enough to finance the climb to $\bar{k}$ would require suppressing the predator and conserving the prey aggressively; but the initial wealth level of the rich is not enough for this to happen. The planner therefore is not able to engineer the escape from the trap.

However, between the blue and green regions lies a more promising case: a band of initial conditions in which laissez-faire and the social optimum *disagree* (orange), with the planner reaching the egalitarian steady state from precisely the configurations where laissez-faire produces the trap. This band captures an initial wealth distribution with intermediate aggregate wealth but high inequality: the rich begin relatively well-capitalized but the poor remain destitute. Under laissez-faire the poor's prey income is eroded as the unmanaged predator boom suppresses the haplochromine stock; the

poor's wealth drifts down, it never reaches $\bar{k}$, and the economy settles into a durable, highly unequal trap in which the rich accumulate far ahead. Under the planner the same poor group is lifted across the threshold: unlike the previous blue region, now that the rich has relatively enough initial capital to invest in Nile perch catch and can obtain enough fishing income, the social planner find it easier to engineer the poor to escape the trap by reducing the booming predator (through transiently encouraging Nile perch catch) and ensuring higher prey income. Interestingly, it is precisely in the most unequal starting configurations that correcting the resource externalities through the social planner changes the long-run distribution.

Panel (B) of Figure 6 reports the aggregate welfare loss of laissez-faire relative to the social optimum across the same initial wealth space, and two patterns stand out. First, the welfare loss is, surprisingly, largest where both groups begin wealthy, i.e., the upper-right region where, as Panel (A) shows, the two regimes nonetheless *agree* on the egalitarian steady state. To understand this, notice first that the welfare gap is a property of the entire transition, not just the destination. When both groups are wealthy, both operate the perch fishery at scale throughout the boom, so the harvesting pressure on the shared stocks, and hence the externality that laissez-faire fails to internalize, is at its greatest. In other words, laissez-faire and the planner end at similar levels of steady state fish stocks, but laissez-faire over-exploits the resource on the way there, and the cost of doing so is largest exactly where catch activity is most intense.

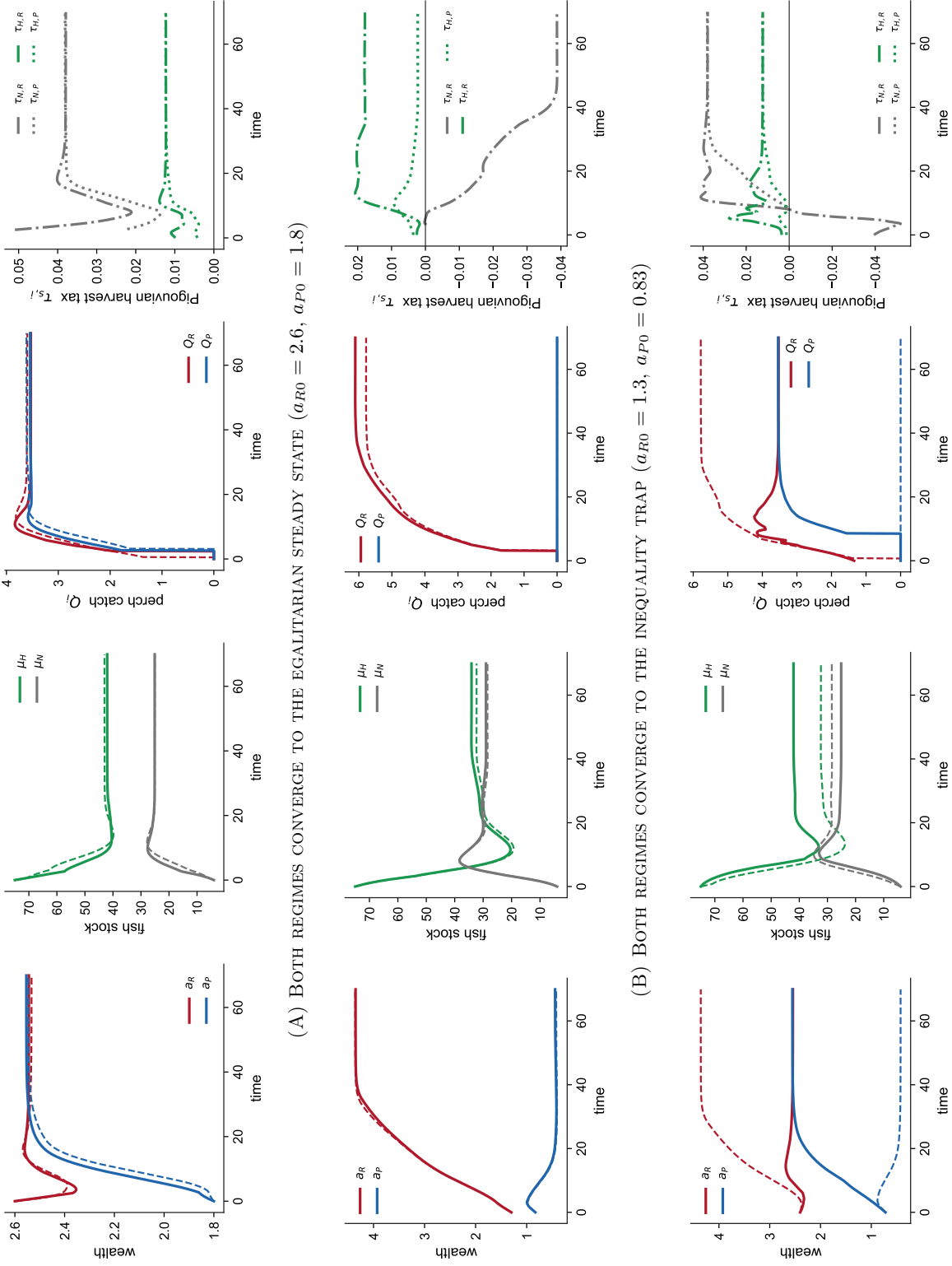
Second, the welfare loss is also substantial in the band where the two regimes disagree. Here the loss is different in kind. It is not the intensive-margin inefficiency of over-harvesting along a shared path, but the extensive-margin cost of the poor getting stuck in the inequality trap: under laissez-faire the poor are permanently excluded from the high-productivity technology and confined to low-income prey harvesting, whereas the planner brings them into the perch fishery. The welfare gap there measures the present value of that foregone escape, i.e., a loss that persists over the entire horizon. Together the two patterns show that laissez-faire is costly for two distinct reasons: it over-exploits the common-pool resource where the Nile perch fisheries activities are most active, and it forecloses the poor's escape from the inequality trap.

4.3 Transition Dynamics and Pigouvian Taxes

To understand how fishing choice evolves endogenously over time, how the stock externality operates before the economy reaches the steady state, as well as how it can be corrected, we now unpack the transition paths. Figure 7 traces three initial conditions, one from each region of the agreement map: both regimes reaching the egalitarian steady state, both reaching the trap, and the disagreement case in which the planner escapes a trap that laissez-faire does not. Throughout, the corrective instrument is the species- and group-specific Pigouvian harvest tax $\tau_{s,i}(x) = W_{\mu_s}(x)/W_{a_i}(x)$ of equation (42): the planner's marginal social value of a stock, expressed as a wedge on the price at which each group harvests it.

Both regimes egalitarian When both groups begin wealthy (Figure 7, top row), the two allocations reach the same egalitarian steady state, but the planner gets there along a less extractive path. Both groups accumulate wealth and take up the perch fishery, and the social optimum brings the poor's wealth up slightly faster: by restraining the common-pool over-harvesting of the shared stocks, it raises the return to the poor's own fishing during the transition. The perch-catch panel makes the mechanism transparent: the planner harvests perch somewhat earlier but, once the fishery matures, holds the catch *below* the laissez-faire level, consistent with conserving a stock it values positively in

FIGURE 7: TRANSITION DYNAMICS UNDER LAISSEZ-FAIRE AND THE SOCIAL OPTIMUM



Notes: Each row traces the transition from the pre-boom initial ecology (\$\mu_{N0}, \mu_{H0} = (4, 75)\$) for a different initial wealth pair, obtained by forward-integrating the optimal feedback rules of each regime. The three rows correspond to the three regions of Figure 6(A): both regimes reach the egalitarian steady state (top), both reach the inequality trap (middle), and the two disagree, with the planner escaping the trap that laissez-faire falls into (bottom). Within each row, solid lines denote the social optimum and dashed lines laissez-faire. The first panel shows wealth (rich \$a_R\$ in red, poor \$a_P\$ in blue, with the dotted line marking the minimum-capital threshold \$k\$ where shown); the second shows the fish stocks (prey \$\mu_H\$ in green, predator \$\mu_N\$ in grey); the third shows the perch catch of each group (\$Q_R\$ in red, \$Q_P\$ in blue). The fourth panel, shown for the social optimum only, plots the Pigouvian harvest taxes \$\tau_{s,i} = W_{s,i}/W_{a_i}\$ that decentralize it, for each species \$s \in \{N, H\}\$ and group \$i \in \{R, P\}\$ (perch taxes in grey; prey taxes in green; rich solid/dash-dot, poor dotted); a positive value is a tax and a negative value a subsidy.

this regime. The implementing taxes are correspondingly both positive: the planner taxes the harvest of *both* species, following directly from equation (42) with $W_{\mu_N}, W_{\mu_H} > 0$, and the tax on the high-value perch is the larger of the two, since it is the more heavily exploited stock whose conservation value is greatest.

Both regimes trapped When both groups begin poor (Figure 7, middle row), the two allocations agree on the inequality trap, and the wealth paths under laissez-faire and the social optimum are accordingly close: the planner does not attempt to engineer an escape from this region. The differences are ecological and distributional. In the long run the planner conserves more of the haplochromine prey on which the trapped poor depend, and slightly more of the predator. The perch-catch panel shows this requires the planner to harvest *more* perch than laissez-faire: since reducing the predator stock relieves predation on the prey, a higher perch catch is the instrument through which the planner protects the haplochromine stock. The implementing taxes reflect this directly: the planner *subsidizes* the perch catch ($\tau_N < 0$, encouraging the rich to thin the predator) while *taxing* the prey catch ($\tau_H > 0$, conserving the stock the poor relies on).

The escape of inequality trap The disagreement case between the social optimum and the laissez-faire is in the bottom row of Figure 7, where the wealth panel shows it starkly: under laissez-faire the poor's wealth drifts down and it remains permanently confined to the prey fishery, while under the social optimum the poor accumulates, crosses $\bar{k}$, and converges toward the rich. Moving to the stock panel, the planner conserves the haplochromine prey aggressively, i.e., holding μ_H well above its laissez-faire level, and to do so it suppresses the predator, so that μ_N under the social optimum runs *below* its laissez-faire level for most of the transition. The perch-catch panel shows how this is achieved through the rich: early on the planner has the rich harvest *more* perch than under laissez-faire, precisely to hold down the predator stock and protect the prey, so that the poor's prey income is high enough to finance accumulation toward $\bar{k}$. Once the poor crosses the threshold and enters the perch fishery itself, the rich's perch catch falls back *below* its laissez-faire level, as the now-shared fishery is harvested with conservation in mind.

The implementing taxes trace the same arc. Throughout, the planner taxes the prey catch ($\tau_H > 0$) to conserve the stock the poor depends on. The perch tax, however, changes sign over the transition: early on it is a *subsidy* ($\tau_N < 0$), encouraging the rich to thin the booming predator while the prey is most depleted and the poor most in need of income; later it becomes a *tax* ($\tau_N > 0$). In other words, while the predator is a distributional liability, i.e., eating the prey that the poor needs to escape, a marginal perch is socially harmful and is subsidized away; once the poor has escaped and the distributional motive lapses, the perch reverts to a valuable harvestable asset and is taxed like any conserved stock. The instrument follows the changing social role of the predator over the transition, exactly as its shadow value changes sign across the two steady states.

4.4 Alternative Policies

The policy instruments of the previous section decentralize the constrained optimum, but they are demanding. Implementing the state-contingent Pigouvian schedule $\tau_{s,i}(x)$ requires the regulator to observe each group's wealth and the ecological state and to levy species- and group-specific harvest taxes that vary continuously with them. This implies an informational and enforcement requirement potentially beyond what fisheries authorities in this setting command. Moreover, the optimum those instruments implement is itself constrained: the planner cannot redistribute wealth and can raise welfare only by reshaping harvest decisions and the stock dynamics they drive. This second observation

cuts two ways. It is what makes the planner’s reach limited, but it also implies that an instrument carrying a *redistributive* channel, which the planner lacks, might raise welfare along a dimension the constrained optimum cannot reach. We therefore evaluate two blunt, widely-debated instruments as policy counterfactuals: a minimum mesh size and a tradable harvest quota. Throughout, we apply each policy to an initial wealth conditions in the inequality-trap region, i.e., the configuration in which laissez-faire implies substantial aggregate welfare loss and a corrective policy is most needed.

4.4.1 Minimum Mesh Size

A minimum-mesh-size regulation requires that a fraction κ of the catch of a targeted species be released. Figure 8 reports its effects when applied to the perch fishery (top row) and to the prey fishery (bottom row).

The first result is that a one-sided mesh regulation on *either* species reduces aggregate welfare, and monotonically so in its stringency (Figures 8A and 8D). The reason is that a mesh size is a blunt quantity restriction: it forces agents to discard part of a catch they have already made, a deadweight loss on the harvesting margin, and it cannot condition on the state or distinguish the externality to make state-contingent conservation. The perch mesh is worse still on ecological grounds. In the inequality trap the predator is precisely the species the planner wants its stock to be reduced, not protected; a perch mesh pushes the ecology in exactly the wrong direction, raising the predator stock and, through the trophic link, depressing the prey on which the poor depend (Figure 8C). The prey mesh moves the ecology in the right direction, i.e., it raises the prey stock (Figure 8F), but achieves this by cutting the effective price the poor receives for its own catch, a first-order income loss to the very group the conservation is meant to help. Either way, the welfare cost dominates.

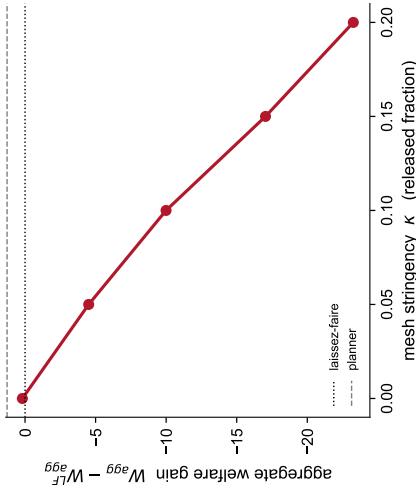
The more striking result is distributional, and the choice of which species to protect reverses the sign of the welfare gap between the rich and the poor. A perch mesh *narrows* the rich–poor welfare gap (Figure 8B), whereas a prey mesh *widens* it (Figure 8E). The mechanism is direct: a mesh on a species acts as a first-order tax (without the lump sum rebate) on whoever harvests it. The perch fishery is operated by the rich, so a perch mesh falls primarily on rich income and compresses the gap; the prey fishery is the poor’s sole activity, so a prey mesh falls on poor income and widens it. The ecological feedbacks, i.e., more predator and less prey under the perch mesh regulation, are present but quantitatively second-order for the distribution relative to this direct incidence. A minimum mesh size is thus welfare-reducing whichever species it protects, and its only material consequence is distributional, working through which group bears such implicit tax the most.

4.4.2 Tradable Quota

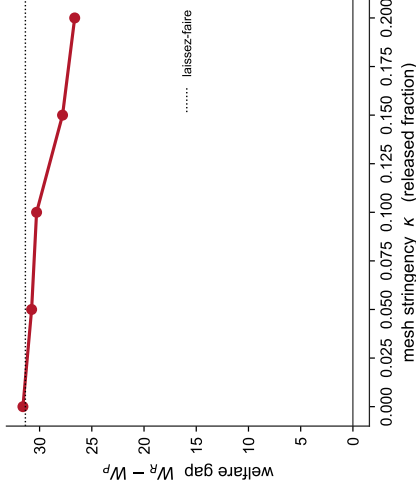
We next consider a tradable quota that caps the aggregate perch catch at a fraction χ of its laissez-faire level, allocates an equal share to each group, and lets permits trade at a market-clearing price p^σ . Because the asset-poor do not operate the perch fishery, they sell their permits to the rich and earn a rent, i.e., a transfer the constrained social planner cannot make.

This redistributive channel substantially changes the welfare picture. Aggregate welfare is *inverted-U*-shaped in the quota (Figure 9A): tightening the cap from no restriction to roughly $\chi = 0.95$ raises welfare above laissez-faire and, over a range of mild quotas, *above the constrained social optimum* itself. However, tightening further reverses the gain and eventually pushes welfare well below laissez-faire. There are two distinctive channels behind this. Like the perch mesh regulation, the quota restricts the perch catch and so protects the predator at the prey’s expense (Figure 9C: a tighter

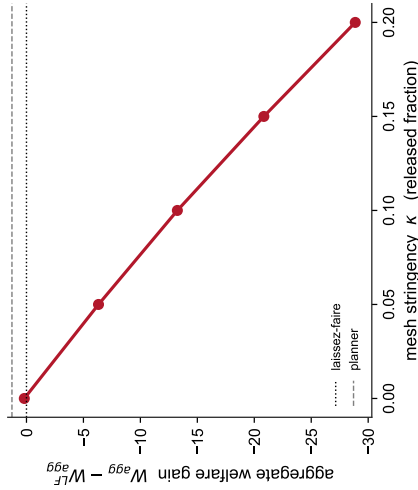
FIGURE 8: MINIMUM MESH SIZE REGULATION



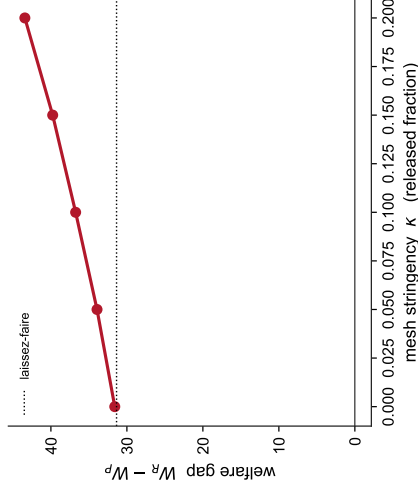
(A) AGGREGATE WELFARE UNDER κ FOR N



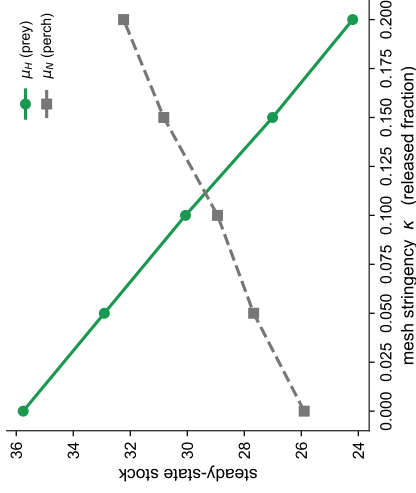
(B) WELFARE GAP $W_R - W_P$ UNDER κ FOR N



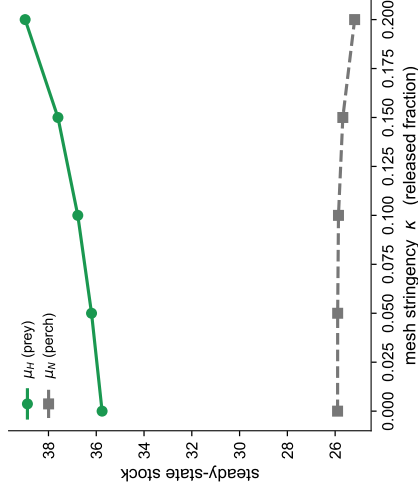
(D) AGGREGATE WELFARE UNDER κ FOR H



(E) WELFARE GAP $W_R - W_P$ UNDER κ FOR H



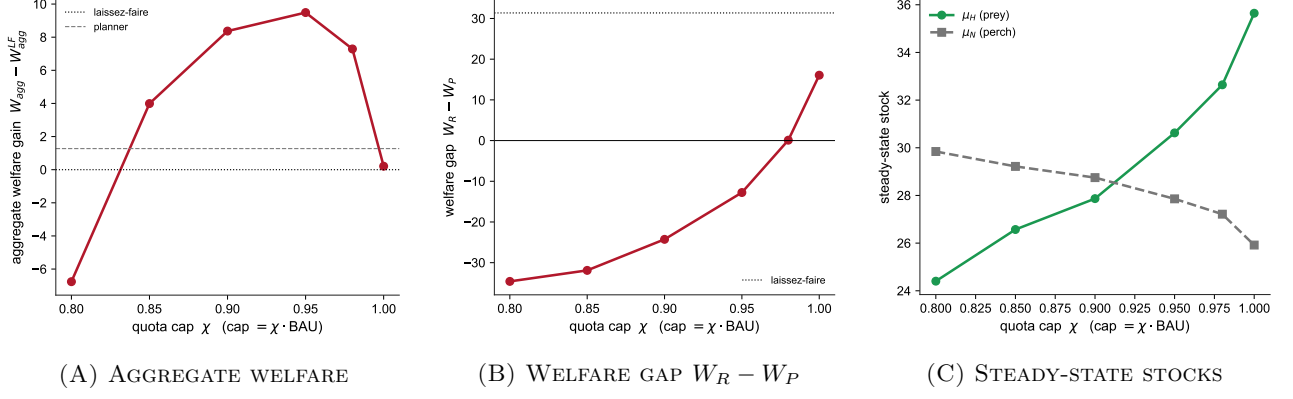
(C) STEADY-STATE STOCKS UNDER κ FOR N



(F) STEADY-STATE STOCKS UNDER κ FOR H

Notes: The figure reports the effect of a minimum-mesh-size regulation that requires a fraction κ of the catch to be released, applied to the perch fishery (top row, panels A–C) or to the prey fishery (bottom row, panels D–F), as the stringency κ increases along the horizontal axis. All counterfactual exercises start from an initial wealth pair in the inequality-trap region under laissez-faire. Each policy is solved as a stationary laissez-faire equilibrium under the regulation and simulated forward to its steady state. The first column reports aggregate welfare relative to laissez-faire, $W_{\text{agg}} - W_{\text{agg}}^{LF}$, with the dashed line at the constrained social optimum. The second column reports the welfare gap between rich and poor, $W_R - W_P$, with the dotted line marking its laissez-faire value. The third column reports the steady-state stocks of prey (μ_H , green) and predator (μ_N , grey).

FIGURE 9: TRADABLE PERCH QUOTA



Notes: The figure reports the effect of a tradable perch quota in which the aggregate catch is capped at a fraction χ of the laissez-faire (business-as-usual) level, $\text{cap} = \chi \cdot \text{BAU}$, so that a lower χ is a tighter quota and $\chi = 1$ imposes no binding restriction. Each agent receives an equal share of the cap and trades permits at a market-clearing price p^σ ; the asset-poor, who do not operate the perch fishery, sell their permits to the rich. The counterfactual exercises start from an initial wealth pair in the inequality-trap region and are solved as stationary equilibria with the permit market clearing at every state. Panel (A) reports aggregate welfare relative to laissez-faire (dotted line) with the constrained social optimum shown for reference (dashed line); panel (B) the welfare gap $W_R - W_P$ (laissez-faire dotted); panel (C) the steady-state stocks of prey (μ_H , green) and predator (μ_N , grey).

quota monotonically raises μ_N and lowers μ_H). But unlike the mesh regulation, the quota also transfers permit rents from the rich to the poor. At mild stringency the redistribution dominates the ecological cost and welfare rises, reaching outcomes the constrained planner cannot, precisely because the planner we modeled earlier is not able to directly make redistributive transfer. At high stringency the prey collapse and the over-restriction of the fishery dominate, and welfare falls. The distributional pattern is monotone and confirms the channel: a tighter quota transfers more rent to the poor, so the rich-poor gap falls steadily and, for sufficiently tight quotas, reverses sign, leaving the poor better off than the rich (Figure 9B).

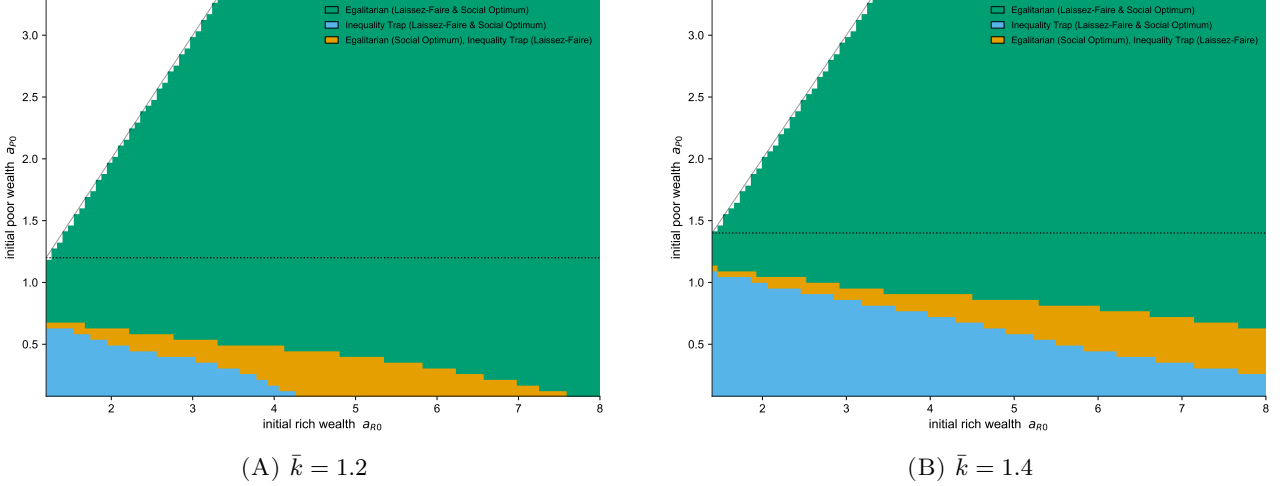
A natural question is whether the poor can use the permit-sale revenue to accumulate past the capital threshold $\bar{k}$ and take up the perch technology, i.e., turning the rent into an escape from the trap. Under our calibration the answer is no. The same tightening that raises the rent also depresses the profitability of the perch fishery, i.e., the permit price p^σ is a cost of operating it, so it simultaneously weakens the incentive to switch into perch. The two forces work against each other, and the poor optimally consumes the rent rather than investing it. Appendix Figure 14 shows this directly: under the quota the poor's consumption rises above its laissez-faire path, but its wealth never approaches $\bar{k}$ and its perch catch remains zero throughout. In other words, the quota raises aggregate welfare and compresses inequality by redistributing resource rents, but it does not help the poor escape the inequality trap.

4.5 Further Results

In this section, we vary the calibrated model by (1) changing the minimum capital requirement $\bar{k}$ to see how this affects steady state selection; and (2) allowing species prices to be endogenous given inverse demand functions.

Alternative $\bar{k}$ Figure 10 presents the results where the left panel is for $\bar{k} = 1.2$ and the right panel is $\bar{k} = 1.4$. Raising the minimum capital requirement enlarges the region (blue and orange) for inequality trap steady state. This is intuitive because higher $\bar{k}$ implies that it is harder for the poor group to take up the Nile perch catch technology through wealth accumulation. The region where the social

FIGURE 10: STEADY STATE WITH ALTERNATIVE MINIMUM-CAPITAL THRESHOLDS



Notes: Each panel reports the steady state equilibrium selected from different initial conditions, varying the minimum-capital threshold $\bar{k}$. The horizontal axis reports the initial wealth of the rich household, a_{R0} , and the vertical axis reports the initial wealth of the poor household, a_{P0} . Panel (A) sets $\bar{k} = 1.2$, while Panel (B) sets $\bar{k} = 1.4$. Colors classify initial conditions according to whether laissez-faire and the constrained social optimum both converge to the egalitarian regime, both converge to the ecological inequality trap, or differ, with the constrained social optimum escaping to the egalitarian regime while laissez-faire remains in the trap. The dotted horizontal line marks the minimum-capital threshold $\bar{k}$, and the solid diagonal line denotes $a_{R0} = a_{P0}$. The regimes are computed by solving the agents' HJB problem on a discretized four-dimensional state space (a_R, a_P, μ_N, μ_H) and classifying the resulting long-run outcomes under laissez-faire and the constrained social optimum.

planner can eliminate inequality trap from the laissez-faire economy (the orange region) also shifts upwards, i.e., now it requires higher initial aggregate wealth (and hence higher initial wealth for both groups) to support the escape.

Endogenous Price The benchmark model treats the prices of Nile perch and haplochromines as fixed. This is useful for isolating the ecological-inequality mechanism, but it shuts down an important general-equilibrium force. In reality, ecological dynamics and wealth accumulation interact through the market system. When a species is harvested intensively, its market supply rises and its price may fall, weakening the incentive to harvest it further. Conversely, when a species is harvested little, scarcity of supply raises its price and strengthens the incentive to harvest it. Endogenizing prices therefore allows the model to capture these price effects.

To incorporate this force, I introduce inverse demand curves for both species. Let $X_{N,t} \equiv Q_{R,t} + Q_{P,t}$ denote aggregate Nile perch catch and $X_{H,t} \equiv q_{R,t} + q_{P,t}$ denote aggregate haplochromine catch. The prices of the two species are determined by isoelastic inverse demand:

$$p_{N,t} = A_N X_{N,t}^{-1/\epsilon_N}, \quad p_{H,t} = A_H X_{H,t}^{-1/\epsilon_H},$$

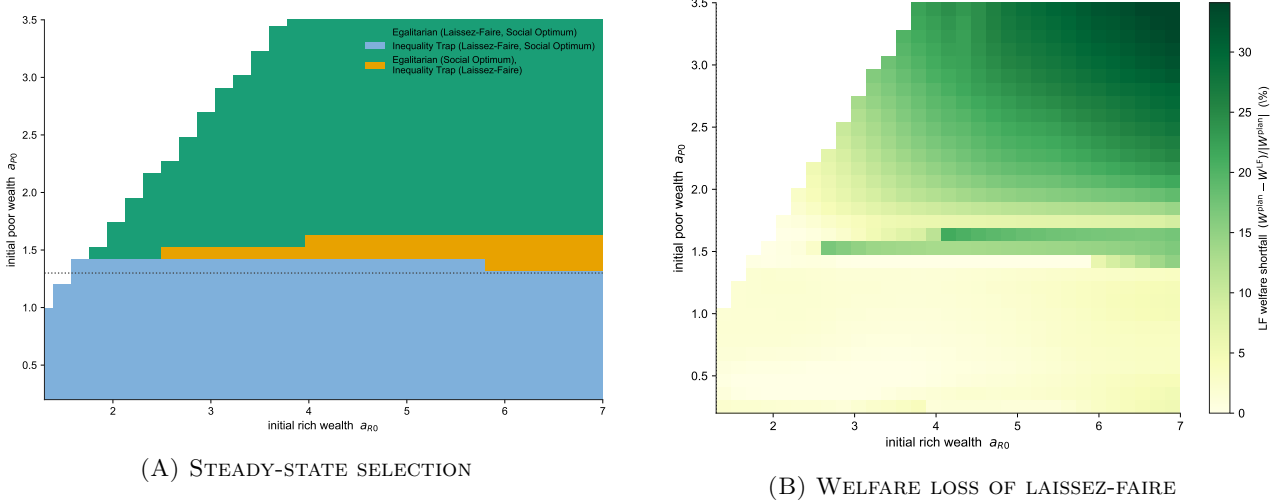
where $A_s > 0$ is a demand shifter and $\epsilon_s > 0$ is the price elasticity of demand for species $s \in \{N, H\}$. A larger ϵ_s implies a more elastic demand curve and therefore a smaller price response to changes in aggregate catch.

To calibrate ϵ_s , we find that [Gallet \(2009\)](#) provide a meta-analysis for estimates of fish price elasticities which are between 0.58-2.02¹³. We choose

$$\epsilon_N = 2.0, \quad \epsilon_H = 1.5.$$

¹³Notice that the fish elasticities vary substantially across products and empirical settings.

FIGURE 11: STEADY-STATE SELECTION AND WELFARE LOSS UNDER ENDOGENOUS PRICES



Notes: Panel (A) reports the long-run regime the economy converges to from each initial condition, obtained by forward-integrating the optimal feedback rules to their rest point and classifying the resulting steady state. Green denotes initial conditions from which both laissez-faire and the social optimum reach the egalitarian steady state; blue, those from which both reach the inequality trap; and orange, those from which the social optimum reaches the egalitarian steady state while laissez-faire is absorbed into the trap. The dotted line marks the minimum-capital threshold $\bar{k}$. Panel (B) reports the welfare loss of laissez-faire relative to the social optimum, $(W^{plan} - W^{LF})/|W^{plan}|$, in percent, where W is discounted lifetime aggregate welfare, defined as the present value of the two groups' flow utilities at equal welfare weights and evaluated along the optimal path from each initial condition. Darker shading indicates a larger welfare loss. Both panels are computed by solving the agents' HJB problem on a discretized four-dimensional state space (a_R, a_P, μ_N, μ_H) and simulating the resulting feedback policies under laissez-faire and the constrained social optimum.

These values place both species in the elastic range of fish demand estimates while preserving the ranking that demand for the lower-value prey species is less elastic than demand for the higher-value Nile perch species. This ranking implies that prey prices respond more strongly to changes in prey catch. This ranking is consistent with the market structure of the Lake Victoria fishery: Nile perch is a high-value species sold through export-oriented supply chains and hence faces a larger market and potentially more substitution possibilities, whereas the lower-value species are more locally consumed. The demand intercepts A_N and A_H are calibrated to match $p_N^* = 0.10$ and $p_H^* = 0.04$ in an inequality trap steady state under laissez-faire. Given $(\epsilon_N, \epsilon_H) = (2.0, 1.5)$, this yields $(A_N, A_H) = (0.26, 0.13)$.

I assume fishing households under both laissez-faire and the social planner's problem remain price takers. Their flow budget constraint becomes

$$\dot{a}_i = p_{N,t}Q_{i,t} + p_{H,t}q_{i,t} - \delta a_i - c_i, \quad i \in \{R, P\}.$$

Thus, prices enter the household problem through market revenues, changing incentives for fishing choices. In particular, I keep the same competitive price-taking market structure in the planner's problem in order to remain comparable with the benchmark model. We abstract away from market power of the social planner and ask how the ecological-inequality interaction changes once species prices are allowed to clear markets.

Panel (A) in Figure 11 reports the steady state selection given initial wealth conditions, replicating Panel (A) in Figure 6 but now allowing for prices to endogenously evolving over time. We find the three types of regions remain: when the two groups begin both rich or both poor, the laissez-faire and the social optimal path will reach the same type of steady state (i.e., egalitarian for the former case, and inequality trap for the latter case). For intermediate aggregate wealth and when the rich

is initially wealthy enough, the social planner will reach egalitarian steady state but the laissez-faire will remain in the inequality trap.

However, it is interesting to note that such disagreement emerges when a_{P0} is slightly higher than $\bar{k}$, i.e., the poor is on the margin to initially be able to catch Nile perch, but they fall back to the labor-intensive technology because of the price effects. To understand this, notice that under laissez-faire, decentralized entry into Nile perch harvesting raises aggregate catch and depresses the perch price, while households do not internalize the ecological effect of harvesting on future stocks. For poor households above but close to the capital threshold, this price and stock feedback makes adoption of capital-intensive technology fragile: the private return to the technology can fall enough that the poor group fails to accumulate wealth and eventually reverts to the inequality trap. The planner avoids this outcome because it internalizes the dynamic ecological externality. The planner preserves the Nile perch stock base in the initial periods and allows the poor to accumulate wealth from Nile perch catch. Thus, even though the poor group is only marginally above the entry threshold, the planner can keep it on a path toward the egalitarian steady state.

This can be seen more clearly in the transition paths of species stock and price in panel (E) and (F) of Figure 12. Nile perch price and stock under the social planner's economy in the initial periods are higher than the laissez-faire, allowing the poor to accumulate wealth by taking advantage of the Nile perch boom when they are on the margin to do so. This is in contrast with panel (C) in Figure 7 where such feature is absent under fixed species prices. However, in later periods, Nile perch (Haplochromines) stock and price under laissez-faire is higher (lower) than the planner's. In other words, under laissez-faire, the economy moves toward a predator-heavy, prey-poor ecosystem which supports a higher price for Nile perch eventually but the poor is excluded. The planner instead supports a more balanced ecological transition, with a larger prey stock and lower long-run Nile perch price, but it allows the poor to remain on the path toward the egalitarian steady state.

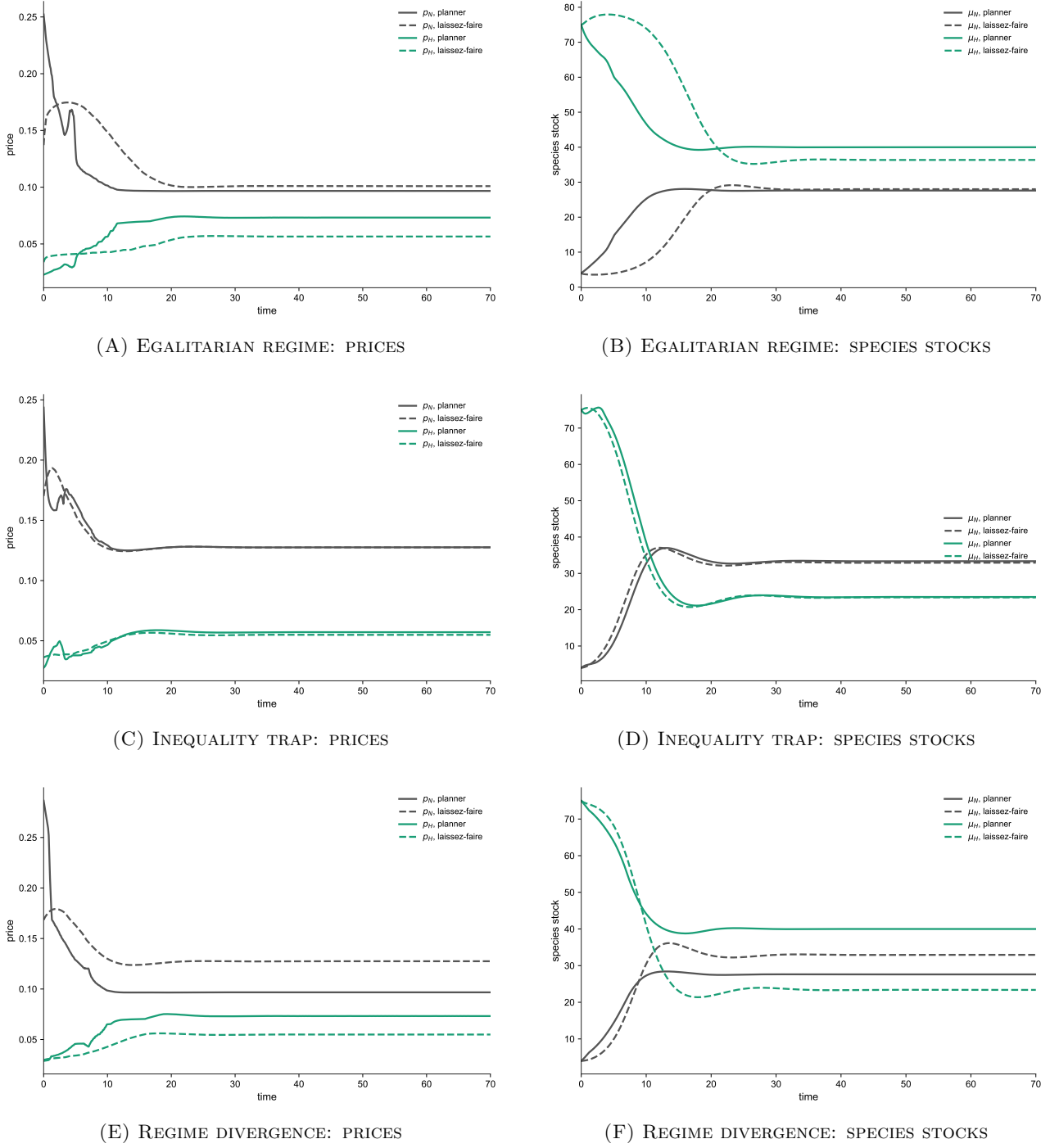
Panel (B) of Figure 11 shows the aggregate welfare loss of laissez-faire relative to the social optimum. We make two observations. First, the regions of initial wealth conditions with substantial welfare loss are similar to that in our benchmark model in Figure 6. These are areas where both groups start with high wealth levels, because the over-fishing problem is the greatest, and where the laissez-faire and social planner disagrees on steady state selection, because of the foregone welfare from escaping the inequality trap. Second, compared to the benchmark welfare loss calculations, the welfare loss with endogenous price is larger in magnitude. This implies that the general equilibrium price effect may amplify the welfare cost of dynamic stock externalities.

Figure 12 also includes the transition dynamics for the other two scenarios where the laissez-faire and the social planner agree on the type of steady state. When both groups begin relatively poor, the social planner and laissez-faire follow similar ecological and price dynamics along the path to the inequality trap steady state. When both groups begin sufficiently wealthy, laissez-faire and the planner both converge to the egalitarian steady state. But the transition paths differ, with the planner inducing faster ecological adjustment, consistent to the corresponding patterns observed in Figure 7.

5 Conclusion

This paper developed a dynamic model in which a predator-prey fishery and the wealth distribution of those who harvest it evolve jointly. Heterogeneous, wealth-constrained agents choose which species to catch and how much to accumulate; a minimum scale of capital in the high-value predator fishery generates a non-convexity, and with it the possibility of an ecological inequality trap in which the

FIGURE 12: TRANSITION DYNAMICS WITH ENDOGENOUS PRICE



Notes: Each panel plots the equilibrium transition dynamics from a representative initial condition (a_R^0, a_P^0) . Rows correspond to three initial conditions. The top row, $(a_R^0, a_P^0) = (4.0, 2.5)$, is one from which both laissez-faire and the social optimum converge to the egalitarian steady state; the middle row, $(4.0, 0.25)$, is one from which both converge to the inequality trap; and the bottom row, $(3.87, 1.48)$, is one from which the social optimum reaches the egalitarian steady state while laissez-faire is absorbed into the trap. The left column reports equilibrium prices and the right column the species stocks. Solid curves denote the social optimum and dashed curves laissez-faire. Grey curves refer to Nile perch (p_N and μ_N); green curves refer to haplochromine prey (p_H and μ_H).

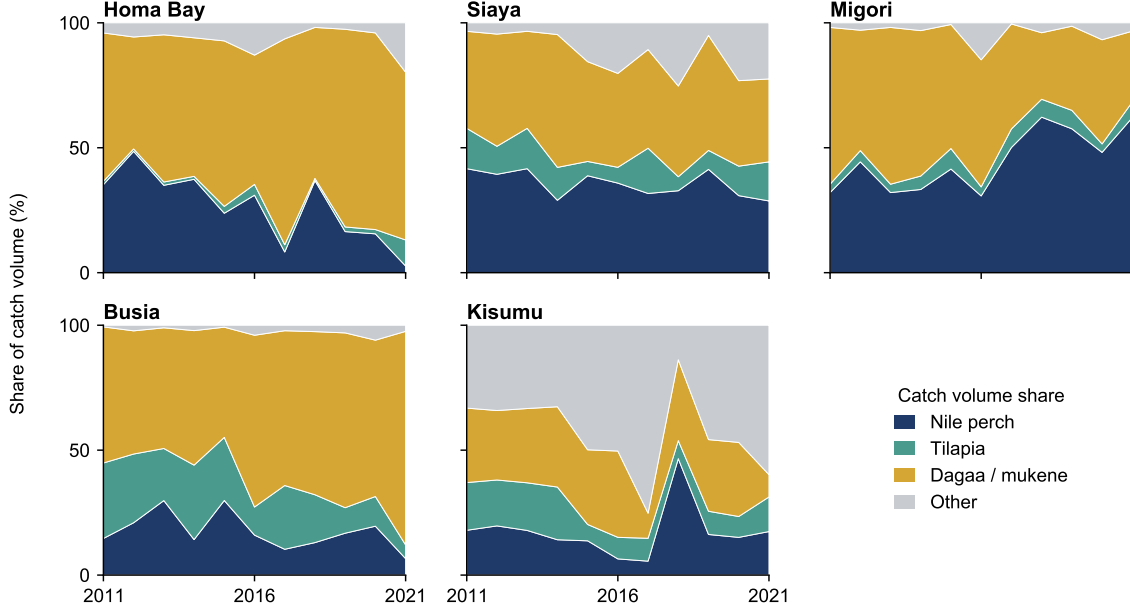
poor are permanently confined to the low-value prey.

Three messages emerge. First, *laissez-faire* is inefficient on two distinct margins: it over-exploits the shared stocks most where economic activity is most intense, and it forecloses the poor's escape into the productive technology. Second, the constrained planner can undo the inequality trap precisely in those initially highly unequal configurations, and can do so without redistributing wealth, purely by repricing the two catches through species- and group-specific Pigouvian taxes, which raise the poor's own income enough to carry it across the capital threshold. Third, the simpler instruments that dominate the policy debate are double-edged: a minimum mesh size works against the very group whose species it protects, while a tradable quota can raise welfare above even the constrained optimum by redistributing resource rents, though those rents do not themselves dissolve the inequality trap. Taken together, the results suggest that in human-ecological systems, conservation and distribution cannot be designed in isolation: which species a policy protects, and through which instrument, determines who bears its cost and who, if anyone, is freed from the trap.

This paper opens up a promising avenue for understanding the distributional consequences of ecological change. Two extensions seem especially worthwhile for future research. First, the ecological side of the model is deliberately stylized: a richer specification incorporating multi-species food webs and stochastic environmental shocks would allow the framework to speak to a wider class of ecosystems and to the resilience of the inequality trap under more realistic dynamics. Second, and more importantly, the central prediction that biodiversity change reshapes the distribution of economic welfare invites empirical scrutiny. Combining measures of the ecosystem using, e.g., satellite data, with georeferenced data on household livelihoods and resource access would make it possible to identify these inequality effects directly, and to empirically evaluate the economic impact of conservation interventions. We view the integration of ecological-economic theory with this emerging body of ecological and economic data as a particularly fertile direction for future work.

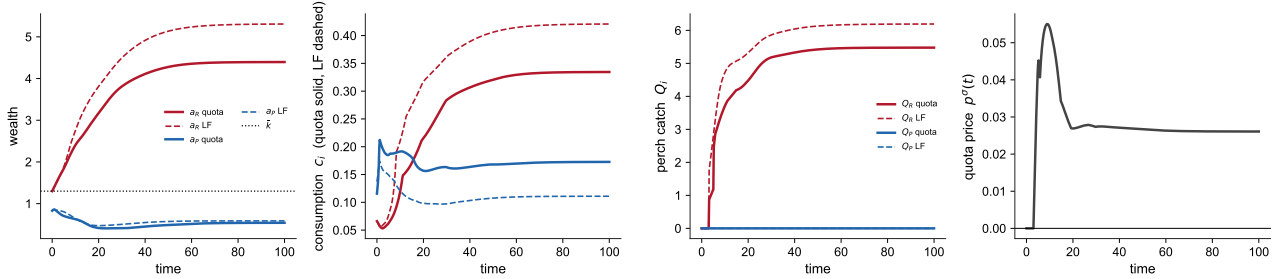
A Additional Figures

FIGURE 13: CATCH COMPOSITION BY COUNTY OVER TIME, KENYA



Notes: Based on Kenya Fisheries Annual Statistical Bulletins, 2011–2021. Each panel corresponds to one of the five Kenyan Lake Victoria counties, ordered by total catch. Within each panel, the stacked areas show the share of each species group in the county’s total catch volume (kg) by year, with species aggregated into four groups: Nile perch (*Lates niloticus*), tilapia (*Oreochromis niloticus* and other tilapias), dagaa/mukene (*Rastrineobola argentea*), and all other species (including haplochromines, catfish, and lungfish).

FIGURE 14: TRANSITION DYNAMICS UNDER A TRADABLE PERCH QUOTA



Notes: Transition paths from an initial wealth pair in the inequality-trap region under a tradable perch quota with cap $\chi = 0.95 \cdot \text{BAU}$ (solid) and under laissez-faire (dashed), over the first 100 periods. The first panel shows wealth (rich a_R red, poor a_P blue; dotted line at the minimum-capital threshold $\bar{k}$); the second consumption c_i ; the third the perch catch Q_i of each group; the fourth the market-clearing permit price $p^\sigma(t)$. Under the quota the poor receives permit-sale income but its wealth remains below $\bar{k}$ and its perch catch stays at zero throughout.

B Derivation

B.1 Static Harvesting Problem

FOCs The capital FOC can be written as

$$p_H \mu_H^{1-\eta} \eta \alpha_1 k_{Hi}^{\eta \alpha_1 - 1} h_{Hi}^{\eta(1-\alpha_1)} = p_N \mu_N^{1-\eta} \eta \alpha_2 \tilde{k}_{Ni}^{\eta \alpha_2 - 1} h_{Ni}^{\eta(1-\alpha_2)}, \quad (45)$$

and the labor FOC as

$$p_H \mu_H^{1-\eta} \eta (1-\alpha_1) k_{Hi}^{\eta \alpha_1} h_{Hi}^{\eta(1-\alpha_1)-1} = p_N \mu_N^{1-\eta} \eta (1-\alpha_2) \tilde{k}_{Ni}^{\eta \alpha_2} h_{Ni}^{\eta(1-\alpha_2)-1}. \quad (46)$$

Dividing (45) by (46) yields

$$\frac{\alpha_1}{1-\alpha_1} \frac{h_{Hi}}{k_{Hi}} = \frac{\alpha_2}{1-\alpha_2} \frac{h_{Ni}}{\tilde{k}_{Ni}}. \quad (47)$$

Capital Threshold Define

$$A_H \equiv p_H \mu_H^{1-\eta}, \quad A_N \equiv p_N \mu_N^{1-\eta}.$$

The H -only branch is

$$Y_H(a; \mu_H) = A_H a^{\eta \alpha_1}. \quad (48)$$

Conditional on operating species N , the group must allocate $\underline{k}$ units of capital to meet the minimum scale requirement. Let $S \equiv a - \underline{k}$ denote the remaining effective capital. For $a \geq \underline{k}$, write $x \in [0, 1]$ for the share of effective capital allocated to species N and $z \in [0, 1]$ for the share of labor allocated to species N . Then

$$Y_M(a; \mu_N, \mu_H) = \max_{x, z \in [0, 1]} \left\{ A_H (1-x)^{\eta \alpha_1} (1-z)^{\eta(1-\alpha_1)} S^{\eta \alpha_1} + A_N x^{\eta \alpha_2} z^{\eta(1-\alpha_2)} S^{\eta \alpha_2} \right\}. \quad (49)$$

For any fixed allocation (x, z) , the ratio of the corresponding N -access payoff to the H -only payoff is

$$\begin{aligned} R_{x,z}(S) &= (1-x)^{\eta \alpha_1} (1-z)^{\eta(1-\alpha_1)} \left(\frac{S}{S + \underline{k}} \right)^{\eta \alpha_1} \\ &\quad + \frac{A_N}{A_H} x^{\eta \alpha_2} z^{\eta(1-\alpha_2)} \frac{S^{\eta \alpha_2}}{(S + \underline{k})^{\eta \alpha_1}}. \end{aligned} \quad (50)$$

Both terms are increasing in S . To see this,

$$\frac{d}{dS} \log \left(\frac{S^{\eta \alpha_2}}{(S + \underline{k})^{\eta \alpha_1}} \right) = \frac{\eta[(\alpha_2 - \alpha_1)S + \alpha_2 \underline{k}]}{S(S + \underline{k})} > 0,$$

where the inequality follows from $\alpha_2 > \alpha_1$. Furthermore, since $R(S)$ is the upper envelope of the functions $\{R_{x,z}(S)\}_{(x,z) \in [0,1]^2}$, it is also strictly increasing. To see this, let (x_1, z_1) be an optimizer at S_1 . For any $S_2 > S_1$,

$$R(S_2) = \max_{x,z} R_{x,z}(S_2) \geq R_{x_1, z_1}(S_2) > R_{x_1, z_1}(S_1) = R(S_1).$$

Hence the ratio

$$R(S) \equiv \frac{Y_M(S + \underline{k}; \mu_N, \mu_H)}{Y_H(S + \underline{k}; \mu_H)}$$

is strictly increasing in S .

At $S = 0$, the N -access branch has no effective capital left after paying the minimum scale cost,

so $Y_M(\underline{k}; \mu_N, \mu_H) = 0 < Y_H(\underline{k}; \mu_H)$. As $S \rightarrow \infty$, choosing the feasible allocation $x = z = 1$ gives

$$Y_M(S + \underline{k}; \mu_N, \mu_H) \geq A_N S^{\eta\alpha_2}.$$

Therefore,

$$\frac{Y_M(S + \underline{k}; \mu_N, \mu_H)}{Y_H(S + \underline{k}; \mu_H)} \geq \frac{A_N S^{\eta\alpha_2}}{A_H(S + \underline{k})^{\eta\alpha_1}} \rightarrow \infty,$$

because $\alpha_2 > \alpha_1$. Hence there exists a unique threshold $a^\dagger(\mu_N, \mu_H) > \underline{k}$ such that

$$Y_H(a; \mu_H) > Y_M(a; \mu_N, \mu_H) \quad \text{for } a < a^\dagger(\mu_N, \mu_H),$$

and

$$Y_M(a; \mu_N, \mu_H) \geq Y_H(a; \mu_H) \quad \text{for } a \geq a^\dagger(\mu_N, \mu_H).$$

B.2 Derivation of $Y_{M,a}$

By the envelope theorem, $Y_{M,a}(a_i; \mu_N, \mu_H)$ equals the common marginal revenue product of capital at the interior optimum. From the first-order conditions for capital allocation across the two harvesting activities, this common value satisfies

$$Y_{M,a} = p_H \eta\alpha_1 \frac{q_i}{k_{Hi}} = p_N \eta\alpha_2 \frac{Q_i}{\tilde{k}_{Ni}}, \quad (51)$$

where $\tilde{k}_{Ni} \equiv k_{Ni} - \underline{k}$ is effective capital in species N . Rearranging each equality gives

$$k_{Hi} = \frac{\eta\alpha_1 p_H q_i}{Y_{M,a}}, \quad \tilde{k}_{Ni} = \frac{\eta\alpha_2 p_N Q_i}{Y_{M,a}}. \quad (52)$$

Summing and imposing the capital-market clearing condition $k_{Hi} + \tilde{k}_{Ni} = a_i - \underline{k}$ yields

$$\frac{\eta(\alpha_1 p_H q_i + \alpha_2 p_N Q_i)}{Y_{M,a}} = a_i - \underline{k}. \quad (53)$$

Define the equilibrium revenue shares

$$s_H \equiv \frac{p_H q_i}{Y_M}, \quad 1 - s_H \equiv \frac{p_N Q_i}{Y_M}, \quad (54)$$

so that $\alpha_1 p_H q_i + \alpha_2 p_N Q_i = [\alpha_1 s_H + \alpha_2 (1 - s_H)] Y_M \equiv \bar{\alpha} Y_M$, where

$$\bar{\alpha}(\mu_N, \mu_H) \equiv \alpha_1 s_H + \alpha_2 (1 - s_H) \in (\alpha_1, \alpha_2) \quad (55)$$

is the revenue-weighted average capital intensity. Substituting and solving for $Y_{M,a}$:

$$Y_{M,a}(a_i; \mu_N, \mu_H) = \bar{\alpha}(\mu_N, \mu_H) \cdot \frac{\eta Y_M(a_i; \mu_N, \mu_H)}{a_i - \underline{k}}. \quad (56)$$

This is the exact multi-sector analogue of the single-technology result $Y_{H,a} = \eta\alpha_1 \cdot Y_H/a_i$: the marginal product of wealth equals the revenue-weighted capital intensity $\bar{\alpha}$ times the average product of *free* capital $Y_M/(a_i - \underline{k})$.

The steady-state Euler condition (26) for the “both-harvest” regime therefore becomes

$$\bar{\alpha}(\mu_N^*, \mu_H^*) \cdot \frac{\eta Y_M(a_A^*; \mu_N^*, \mu_H^*)}{a_A^* - \underline{k}} = \rho + \delta. \quad (57)$$

Although no fully explicit closed form for a_A^* exists (since $\bar{\alpha}$ and Y_M themselves depend on a_A^* and the equilibrium species stocks), (57) characterises the steady state sharply and mirrors the structure of (30) for the poor.

B.3 Binding Capital Constraint in Social Optimum

Let $\lambda_i \geq 0$ denote the multiplier on the productive-capital constraint in (8). Conditional on the chosen regime, the capital allocation satisfies the Kuhn–Tucker conditions

$$\mathcal{P}_{Hi}(x)q_k(k_{Hi}, h_{Hi}; \mu_H) - \lambda_i \leq 0, \quad k_{Hi} \geq 0,$$

$$k_{Hi} [\mathcal{P}_{Hi}(x)q_k(k_{Hi}, h_{Hi}; \mu_H) - \lambda_i] = 0,$$

and

$$\mathcal{P}_{Ni}(x)Q_k(k_{Ni}, h_{Ni}; \mu_N) - \lambda_i \leq 0, \quad k_{Ni} \geq 0,$$

$$k_{Ni} [\mathcal{P}_{Ni}(x)Q_k(k_{Ni}, h_{Ni}; \mu_N) - \lambda_i] = 0,$$

together with

$$\lambda_i(a_i - k_{Hi} - k_{Ni}) = 0.$$

Thus the productive-capital constraint binds whenever at least one operated species has strictly positive social marginal catch value. In that case, leaving productive wealth idle cannot be optimal, since reallocating idle capital to that operated species strictly raises the planner’s objective.

C Data Construction

The data were obtained from the Demographic and Health Surveys Program (DHS), which provides nationally representative household surveys covering population, health, and nutrition indicators across developing countries. For Kenya, the survey waves included 1989, 1993, 1998, 2003, 2008, 2014, and 2022. For Tanzania, the included waves were 1996, 1999, 2003, 2004, 2007, 2010, 2011, 2015, and 2022. For Uganda, the included waves were 1995, 2000, 2006, 2011, and 2016. These country-year datasets were processed separately before being appended into three country-specific datasets. Because DHS surveys use standardized questionnaires but may vary in country-specific coding schemes and survey-wave variable availability, variables were standardized within each country-year dataset using the corresponding DHS variable labels and value labels.

The sample was selected by region and occupation. First, observations were limited to regions closely connected to the Lake Victoria zone. These regions were identified through Google Maps and then matched to the regional classifications available in the DHS datasets. For Uganda, the retained DHS regions were Central, Kampala, East Central, South Buganda, and Busoga. For Tanzania, the retained regions were Kagera, Mwanza, Mara, Simiyu, and Geita. For Kenya, the retained regions were Busia, Siaya, Kisumu, Homa Bay, Migori, Lamu, Garissa, and Wajir. Second, within the selected regions, the sample was further restricted to households in which either the respondent or the respondent’s partner reported an occupation related to fishing or fishery-adjacent activities. Because occupational classifications are not fully consistent across countries and survey waves, occupation codes were manually reviewed and harmonized according to occupational content. The standardized occupational classification consists of four broad categories: skilled fishery workers, subsistence or mixed fishery workers, fishing-related support and transport workers, and broad agriculture-related workers.

The processed dataset also kept variables measuring household living standards, household assets, and individual-level demographic and health conditions. Household living standards are measured by housing materials, including wall, floor, and roof materials, and by access to basic services such as electricity, toilet facilities, and drinking water sources. Household assets are measured by ownership of durable goods, including a television, radio, refrigerator, bicycle, motorcycle, car, and telephone. Individual-level demographic and health conditions include child mortality and dietary composition, women’s height and weight, marital status, age at first marriage or cohabitation, age at first birth, number of children ever born, education, and spouse’s education.

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